

The Relationship Between Plasma and Tissue Lipids in Human Atherosclerosis

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I. Introduction

A high level of plasma cholesterol has emerged as an important risk factor in all epidemiological studies on the incidence and prevalence of ischemic heart disease. However, in man the relationship between plasma cholesterol levels and the extent and severity of atherosclerosis and the cholesterol content of the arterial wall has not emerged clearly. This probably arises from at least two causes. First, it is difficult to obtain information on plasma cholesterol levels during life in subjects coming

to autopsy; consequently, the few studies that are published are based on institutionalized patients of advanced age (Paterson *et al.*, 1956, 1963); second, the atherosclerotic lesions found in arteries of middle-aged or older humans are extremely heterogeneous, ranging from accumulation of lipid, particularly cholesterol ester, with very little increase in connective tissue components to a massive hyperplasia of collagen and smooth muscle cells with only slight accumulation of lipid. The balance between the different types of lesion is clearly influenced by many factors including age, sex, and race (McGill, 1968), but the influence of plasma lipid levels is not clear.

There is, however, a close correspondence between the detailed lipid composition of the S_{10} 0-12 (LD) lipoprotein of plasma and the composition of extracellular lipid in samples of intima with defined histological characteristics (Smith *et al.*, 1967, 1968; Smith and Slater, 1972a). There is also a substantial amount of electrophoretically and immunologically intact LD-lipoprotein in the human aortic intima, and its concentration in normal intima is highly correlated with the level of the subject's plasma cholesterol during the week before death (Smith and Slater, 1972b, 1973a,b). From the analysis of the cholesterol esters in topographically and morphologically defined fractions isolated by microdissection, Smith and Slater (1972a) concluded that most of the cholesterol in the main "atheroma" lipid pool of large plaques in human aorta is derived directly from plasma LD-lipoprotein and not from disintegration of fat-filled cells. This suggests that in humans, the entry of LD-lipoprotein, its retention within the intima and the precipitation of lipid from it may be key factors in atherogenesis.

This is the aspect of the problem which will be discussed in this paper; several reviews on other aspects of arterial lipids have already appeared in *Advances in Lipid Research* (Clarkson, 1963; Zemplényi, 1964; Day, 1967; Wissler and Vesselinovitch, 1968; Adams, 1969; Whereat, 1971), and in particular Portman (1970) has published a most comprehensive review on esterified fatty acids and cholesterol. No attempt will be made to cover the same ground as these papers or to emulate their completeness.

II. Relationship Between the Chemical and Morphological Characteristics of Lipid in Human Aortic Intima

A. MORPHOLOGY OF NORMAL INTIMA AND ATHEROSCLEROTIC LESIONS

Much of the confusion and apparent disagreement in atherosclerosis research has undoubtedly arisen either from failure to differentiate be-

tween different types of lesion or because the same name has been applied to tissue samples which are not in fact comparable. In the hope of avoiding further confusion and repetitive descriptions in the text the morphological characteristics of the intimal samples are briefly described below.

1. Normal Intima

In children of the first decade and in virtually all experimental animals the normal intima consists of a single layer of endothelial cells separated from the internal elastic lamina by a very thin subendothelial layer containing small amounts of fine collagen and elastic fibers and glycosaminoglycans (GAG). In man by the third decade, the subendothelial layer is invariably thickened, and contains several layers of smooth muscle cells and bundles of collagen fibers. This "diffuse intimal thickening" increases progressively with age; its significance was recently reviewed by Geer and Haust (1972). All samples of "normal" intima in adult humans have some degree of diffuse intimal thickening, and this is an important difference from normal intima in experimental animals. From the fourth decade onward, two upper (luminal) layers can be distinguished in the aorta—a superficial layer containing collagen and smooth muscle cells in which elastin cannot be demonstrated by the Verhoeff stain and a deeper layer which lies between this and the first intact medial elastic lamina and contains numerous fragmentary elastic laminae (Taylor, 1953; Smith, 1974). It is not clear whether this layer should be regarded as intima or media. In young subjects, very little lipid can be demonstrated by sudan staining, but from about age 30 onward there is an increasing accumulation of fine, extracellular, perifibrous, sudanophilic droplets. These have a diameter of 0.5–1.5 μ and seem to be particularly associated with the fragmentary elastic laminae in the deep layer of the intima (Smith *et al.*, 1967).

2. Fatty Infiltration

Arteries stained macroscopically for fat (Holman *et al.*, 1958) generally show areas of intense, focal staining corresponding with fatty streaks, nodules, or plaques. They may also show patches of more diffuse sudanophilia that correspond to microscopic collections of extracellular lipid droplets, slightly coarser than the perifibrous lipid droplets of normal intima, which are scattered through an otherwise rather normal intima; fine sudanophilic droplets may also be found in the smooth muscle cells. These areas of "fatty infiltration," which can only be detected in unstained arteries if the lipid infiltration is unusually massive, probably occur more commonly in the coronary arteries than in aorta; they are frequently described as "fatty streaks," but microscopically

their morphology is quite different from the lesion containing numerous fat-filled cells which is the typical aortic fatty streak of the second decade. The confusion arising from using the same term for lesions of different morphology, and probably also different etiology, is reflected in the controversy as to whether the fatty streak is the precursor of the fibrous plaque.

3. *Fatty Streaks*

Macroscopically, in unstained arteries, typical juvenile fatty streaks appear as slightly raised whitish or pale yellow spots, 2–3 mm in diameter, and generally occurring in clusters. In the aorta, they frequently occur in lines between the orifices of the intercostal arteries. They are found at all ages after birth, reaching a maximum extent about age 20 (Holman, 1961). Microscopically, the lesions contain clusters of cells filled with *large* lipid droplets. These fat-filled cells may be confluent, or slightly separated from each other; they may occur at any depth in the intima but are most frequent in the superficial (luminal) layers and are found rarely, if at all, in aortic media. In the human, they are probably mainly smooth muscle cells and “ovoid foam cells” (Geer and Haust, 1972) of uncertain origin. In the author’s opinion, fatty streaks must be defined on the microscopic basis that most of the lipid is within fat-filled cells.

4. *Raised Fatty Nodules and Fatty Plaques*

In larger fatty streaks and nodules, extracellular lipid may accumulate between and beneath the fat-filled cells, and this process is probably extended to produce macroscopically raised, yellow “fatty” plaques. Microscopically there is a thin “cap” consisting mainly of fat-filled cells overlying an “atheroma pool” of amorphous, extracellular lipid. It appears that in the human aorta this type of structure is not commonly found in large lesions. Smith and Slater (1972a) found that in larger macroscopically “fatty” plaques, the fat-filled cells were generally embedded in a fairly thick, fibrous cap overlying the atheroma lipid center.

5. *Gelatinous Thickenings and Plaques*

Macroscopically, these lesions have a characteristic translucent appearance, often slightly pinkish, and on stripping or cutting the intima has a “gelatinous” texture. Microscopically, they are very ill-defined, the only consistent finding being a loose structure with wide gaps between the smooth cells and rather thick collagen bundles. Lipid is variable, ranging from virtually no staining, through intense diffuse sudanophilia or heavy perifibrous lipid to accumulation of extracellular “atheroma” lipid

in deep layers. It seems probable that these lesions are the precursors of fibrous plaques; fibrous plaques frequently have gelatinous peripheries that seem to be their growing points. The literature on this type of lesion has recently been reviewed by Haust (1971) and Geer and Haust (1972). They appear to correspond to the "focal intimal edema" described by German workers, or the insudation lesions described for example, by More and Haust (1957).

6. Fibrous Plaques

These are sharply raised lesions that macroscopically appear pearly white in both stained and unstained arteries. Microscopically, there is a thick, compact, collagen cap that may show very little lipid staining or patches of normal or abundant perifibrous lipid, generally in the deeper layers; occasionally, there are scattered smooth muscle cells containing a few fat droplets or Nile-blue-positive granules (Smith and Slater, 1972a). In some fibrous plaques, lipid staining is low throughout the whole depth, but more characteristically, there is a pool of amorphous atheroma lipid in the deep layers. This consists of a mass of fine, extra-cellular lipid droplets or granules, which may be interspersed with coarser droplets and "cholesterol crystals," and which extends down to the internal elastic lamina and sometimes into the upper media. Plaques complicated by ulceration, calcification, hemorrhage, or thrombosis will not be discussed in this paper.

B. CONCENTRATION AND COMPOSITION OF THE LIPIDS

The amount and composition of lipid differs greatly between intima and media, and between normal intima and the different types of lesion described above; consequently, only studies on defined tissue samples will be discussed.

1. Normal Intima

In normal intima there is a progressive increase in the concentration of most lipid classes with age (Smith, 1965a) and morphologically this is correlated with increasing numbers of perifibrous lipid droplets that are particularly, but not exclusively, associated with fragmentary elastic laminae in the deeper layers (Smith *et al.*, 1967). Total phospholipid, free cholesterol, and triglyceride increase very slowly, approximately 0.2 mg/100 mg defatted dry tissue per decade for each group, but from about age 20 upward, cholesterol ester increases at about 5 times this rate. Thus, although cholesterol ester is a minor component of normal intimal lipid up to about age 30, it becomes the main component above age 40.

The original data suggested that cholesterol ester was very low until the rapid increase starts in the second decade, and this has now been examined in more detail; results on aortic samples from 32 subjects aged 6 months to 39 years, in which at least 60% of the tissue appeared to be intima, are shown in Fig. 1. Ester cholesterol remains almost constant at a level of about 0.4 mg/100 mg defatted dry tissue over the first two decades, but over the third and fourth decades it rises steeply ($r = 0.863$, $p \ll 0.001$) at a rate of 0.63 mg per decade. This is in contrast to free cholesterol which rises continuously over the whole period at a rate of 0.17 mg per decade ($p < 0.001$). The ratio free cholesterol/phospholipid shows no significant change from a mean value of 0.45 ± 0.11 , which is very close to a 1:1 molar ratio; this relationship seems to be maintained up to age 60. This suggests that free cholesterol and phospholipid are linked, either in the membranes of the cell, or in a liquid

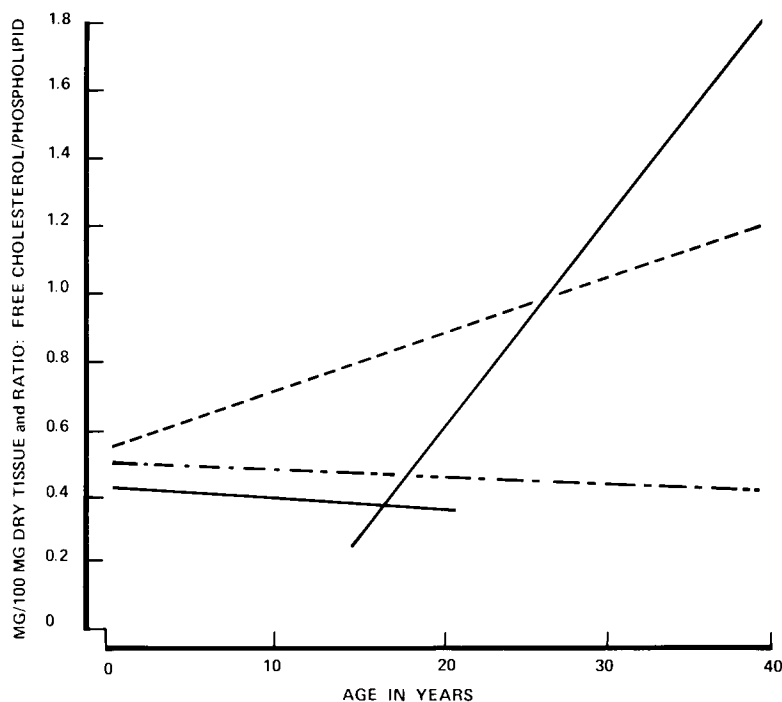


FIG. 1. Regression lines for the relation between free and esterified cholesterol concentrations and age in normal intima from 32 subjects, aged 6 months to 39 years. For esterified cholesterol, separate regressions have been calculated for 6 months to 20 years and 15 to 39 years. The ratio of free cholesterol/phospholipid is also shown. (Esterified cholesterol —; free cholesterol — — —; ratio: free cholesterol/phospholipid — · — · — ·.)

crystalline lattice (Small, 1970). By contrast, the increase in ester cholesterol, starting in the 15–20 age range seems to be an independent process, which may be related to the amount of intact plasma low density (LD) lipoprotein in the intima, and this will be discussed later. Eisenberg *et al.* (1969) found a continuous increase in cholesterol ester in preparations that included most of the media; in the young age-groups, the contribution of the intima would be negligible. In ascending aorta and arch from children aged 1 month to 10 years, Day and Wahlqvist (1970) found comparable concentrations of free cholesterol, but much less ester cholesterol. Their published photograph suggests that their preparations contained at least 80% media, and the ester cholesterol concentration is similar to that found in intima-media preparations in this laboratory.

In addition to the rapid increase in the concentration of cholesterol ester with age, there is a change in the cholesterol ester fatty acid pattern, so that with increasing age the whole composition of normal intimal lipid approaches the composition of the LD-lipoprotein of adult plasma (Smith, 1965; Smith *et al.*, 1967). Intimal lipid composition in three age groups is compared with isolated plasma S_r 0–12 lipoprotein in Table I. The cholesterol esters in the youngest age groups are particularly characterized by a high proportion of C_{16} acids and a low proportion of 18:2 (linoleic) acid; there are also higher proportions of both long and short chain minor components, and of 18:0 (stearic) acid. These findings are in close agreement with Day and Wahlqvist (1970), and with the findings of Portman and Alexander (1966) on Rhesus monkeys. It appears that this type of fatty acid pattern is also characteristic of the serum cholesterol esters in newborn and very young children (Zöllner *et al.*, 1966; Zee, 1968). Day and Wahlqvist (1970) found a high ratio of 18:1/18:2 acids up to 1 year, but in the 1–5 year age-group the ratio fell to unity, and in the 5–15 age-group reached the adult level of 0.75; the “intimal” ratio also fell, and between 1 and 10 years of age, it was about twice the plasma ratio. The authors conclude that all this cholesterol ester is derived from plasma.

Our results seem more complex—the constancy of the ester cholesterol concentration during the first two decades suggest that this may represent “endogenous” cholesterol ester and that plasma cholesterol ester does not accumulate until about age 20 when the concentration starts its steep and linear rise (Fig. 1). If this is correct, the cholesterol ester fatty acid pattern might be expected to remain constant over the first two decades, when the concentration remains constant, and then to change rapidly toward the pattern found in serum when the concentration starts to rise. However, using the proportion of linoleic acid (18:2) in the

Table I
COMPOSITION OF THE LIPID IN NORMAL INTIMA (Perifibrous Lipid Only)

No. of samples	Intima with perifibrous lipid				Plasma S_t 0-12 lipoprotein
	Age 6 mo.-20 yr	Age 40-59	Age over 70		
Total lipid (mg/100 mg tissue) ^a	10	33	13	22	—
	4.3	10.8	16.0		
Percentage of total lipid					
Cholesterol ester	16.7 ± 6.8 ^c	42.1 ± 5.7	57.2 ± 7.9	58.2 ± 2.2	
Free cholesterol	17.6 ± 3.7	13.0 ± 2.3	10.5 ± 1.1	11.5 ± 1.0	
Phospholipid	42.7 ± 7.5	29.3 ± 3.6	17.2 ± 4.8	20.0 ± 1.5	
Triglyceride	23.0 ± 10.5	15.4 ± 6.1	15.1 ± 5.7	10.2 ± 2.6	
CEFA ^b					
Percentage 18:2 in 18:1 + 18:2 fraction	39.5	58.0	58.7	66.0	
Percentage of total					
14:0	3.4 ± 2.1	1.7 ± 0.8	1.2 ± 0.7	1.0 ± 0.3	
16:0	23.6 ± 6.7	14.5 ± 1.5	13.7 ± 1.7	13.7 ± 3.2	
16:1	9.4 ± 2.7	5.5 ± 2.5	6.3 ± 1.8	4.2 ± 0.6	
18:0	7.1 ± 3.9	1.5 ± 1.8	0.8 ± 0.5	2.5 ± 1.0	
18:1	27.2 ± 6.5	28.0 ± 4.4	27.4 ± 4.5	24.1 ± 3.1	
18:2	17.8 ± 9.9	38.6 ± 7.0	40.0 ± 7.3	46.8 ± 6.6	
20:3	0.7 ± 0.6	1.0 ± 0.5	0.9 ± 0.4	trace	
20:4	3.3 ± 2.3	5.3 ± 1.0	5.2 ± 1.2	5.2 ± 1.2	
Minors	7.1	3.9	4.5	2.5	
Ratio: free cholesterol/phospholipid	0.41	0.44	0.61	0.57	

^a Defatted dry tissue.

^b Cholesterol ester fatty acid.

^c ± Standard deviation.

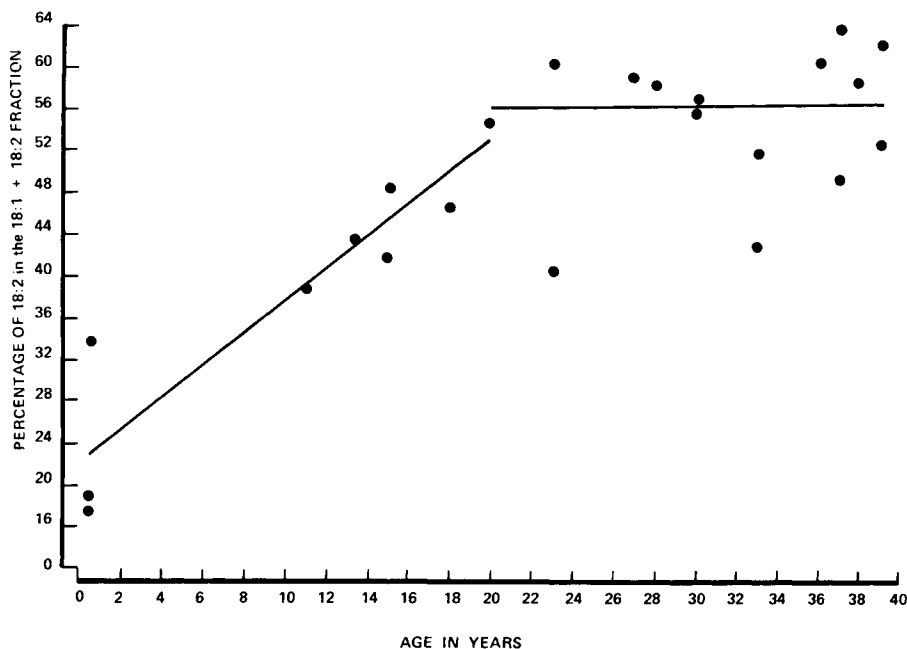


FIG. 2. Relation between age and the percentage of linoleic (18:2) acid in the 18:1 + 18:2 fraction of the cholesterol ester fatty acids from normal intima (23 subjects, aged 6 months to 39 years).

18:1 + 18:2 fraction as an index, it can be seen in Fig. 2 that the reverse happens: the percentage of 18:2 rises rapidly during the first two decades when the concentration is constant, but between ages 20 and 39 it appears to have reached a maximum of $56.5 \pm 5.8\%$ linoleic acid and shows no significant increase in older age groups although the concentration of cholesterol ester increases rapidly. Thus during the first two decades, the arterial cholesterol ester seems to be in equilibrium with the plasma cholesterol ester, although there is no net accumulation. If this amount of cholesterol ester were present in the form of contaminating or reversibly-bound plasma lipoprotein, it could be measured by quantitative immunoelectrophoresis (Smith and Slater, 1972b), but in the first decade we have failed to detect any intact LD-lipoprotein, and in the second decade, the amount of lipoprotein found could account for only about one-third of the ester cholesterol.

Total phospholipid increases linearly with age in normal intima. Smith (1965a) found no significant change in the ratio lecithin/sphingomyelin with age in intima, but in media there was a marked increase in the

proportion of sphingomyelin between ages 15 and 55. The use of intima + media preparations probably accounts for the continuous increase in the proportion of sphingomyelin with age found by Eisenberg *et al.* (1969). These authors find that aortic lecithinase increases about 5-fold between the first and tenth decades, whereas sphingomyelinase decreases slightly; it seems important that this work should be extended to pure intimal preparations where the proportion of sphingomyelin does not apparently increase, and the tissue is in contact with substantial amounts of plasma LD-lipoprotein. Arterial phospholipids were expertly reviewed by Portman in 1970.

Triglycerides also increase slowly with age in normal intima (Table I) but in lesions they generally increase less than the other lipid components (Tables I, II, and VI; Böttcher, 1961; Insull and Barch, 1966) and consequently have not aroused much interest. Earlier reports that they occur in significantly higher concentration in coronary arteries than in aorta (Böttcher *et al.*, 1960; Böttcher, 1961) have not been confirmed (Abdulla *et al.*, 1969; Panganamala *et al.*, 1974).

2. Fatty Infiltration

There appears to be little information on this rather ill-defined lesion which may resemble a fatty streak by macroscopic staining. Day and Wahlqvist (1970) compared normal with lesion in children aged 1 month to 10 years and the only consistent difference was in the percentage of the cholesterol esterified; the lesion contained approximately twice the proportion of esterified cholesterol in all age groups. In this laboratory, 19 samples containing a mixture of abundant perifibrous lipid and occasional scattered cells containing fat droplets showed an increase of 44% in total cholesterol, compared with normal intima and a cholesterol ester fatty acid pattern intermediate between normal intima and fatty streaks.

3. Fatty Streaks (as defined in Section II,A,3)

Typically, lesions in which most of the lipid is within fat-filled cells contain a high proportion of cholesterol ester of very characteristic fatty-acid pattern (Smith, 1965a; Geer and Guidry, 1964; Smith *et al.*, 1967). Lipid patterns in fatty streaks in which the numerous fat-filled cells are either scattered or in confluent groups are shown in Table II, where they are compared with fat-filled cells isolated by microdissection from the caps of large atheromatous plaques. The maximum divergence from the plasma pattern occurs in the fatty streaks containing groups of confluent fat-filled cells. These lesions were found in a remarkably restricted age-group; 8 of the 10 subjects were aged 38–47 and the remaining two subjects were aged 19 and 51. There is no significant difference in the

Table II
COMPOSITION OF THE LIPID IN FATTY STREAKS AND ISOLATED FAT-FILLED CELLS

	Fatty streaks			Fat-filled cells from caps of plaques	
	Scattered fat-filled cells	Confluent fat-filled cells	Small and scattered	Large and confluent	
No. of samples	15	10	8	21	
Total lipid (mg/100 mg tissue)	38.7 ± 7.2	40.0 ± 8.7	45.1 ± 7.4	115.6 ± 15.6	
Percentage of total lipid					
Cholesterol ester	64.1 ± 7.6	68.7 ± 5.8	62.9 ± 3.5	65.9 ± 2.6	
Free cholesterol	13.1 ± 2.9	10.1 ± 2.9	16.5 ± 2.0	14.1 ± 1.4	
Phospholipid	16.2 ± 3.0	14.8 ± 2.2	14.6 ± 2.0	13.4 ± 1.4	
Triglyceride	6.6 ± 1.1	6.4 ± 2.7	7.2 ± 1.4	6.5 ± 1.6	
CEFA					
Percentage 18:2 in 18:1 + 18:2 fraction	24.4 ± 5.1	18.6 ± 3.5	24.3 ± 8.8	29.6 ± 6.8	
Percentage of total					
16:0	13.6 ± 3.3	10.7 ± 3.1	With the very small samples obtained from microdissection the 14:0 to 18:0 and 20:3 and 20:4 acids are not reliable		
16:1	6.7 ± 1.4	7.7 ± 1.5			
18:0	2.5 ± 1.5	2.0 ± 1.6			
18:1	47.6 ± 5.3	52.4 ± 4.3			
18:2	15.2 ± 4.5	12.1 ± 3.0			
20:3	4.0 ± 1.6	5.0 ± 1.7			
20:4	3.9 ± 1.9	3.6 ± 1.4			
Ratio: free cholesterol/phospholipid	0.81	0.68	1.1	1.05	

composition of the lipids in fatty streaks and fat-filled cells dissected from the caps of large plaques.

The ratio free cholesterol/phospholipid is approximately twice the ratio in normal intima, but there appears to be no significant difference in the composition of the phospholipids from fatty streaks and adjacent normal intima (Smith, 1965a). Lang and Insull (1970) have divided fatty streak lipids into droplets and residue; 95% of the droplet lipid was cholesterol ester, and the ratio free cholesterol/phospholipid was 1.7, approximately 4 times the ratio in normal intima, but in the residue, the ratio was close to that for normal intima. The fatty acid composition of the droplet cholesterol esters was very similar to the fatty streaks with confluent fat-filled cells shown in Table II. The composition of the phospholipids in both fractions was rather markedly different from that found by Smith, but the proportion of sphingomyelin in the droplets was significantly lower than in the residue.

4. *Small Raised Fatty Nodules and Plaques*

The lipid composition is highly correlated with the detailed morphology of the lesion (Smith *et al.*, 1968). With increasing amounts of amorphous atheroma lipid in the deep layers of the lesion, there is a marked increase in the proportion of free cholesterol, decrease in proportion of ester cholesterol, and change in the cholesterol ester fatty acid pattern. These changes are illustrated in Table III, which shows the lipid composition in a series of small lesions obtained from a single aorta. The concomitant increases in the amount of amorphous lipid, in the proportion of free cholesterol, and in the proportion of cholesterol linoleate suggest that formation of the amorphous lipid may involve breakdown of fat-filled cells with the release of cholesterol esterases that catalyze preferential hydrolysis of cholesterol oleate.

In an attempt to demonstrate hydrolysis we incubated minced or homogenized intima for periods between 14 and 20 hours and compared the proportions of free cholesterol and of oleic and linoleic acids in the cholesterol ester fatty acids in control (frozen) and incubated samples (E. B. Smith and R. S. Slater, unpublished). An increase in free cholesterol greater than 5% occurred in only 8 of 42 tissue samples (19%), comprising no samples of normal intima, 3 of 9 fatty streaks and nodules (33.3%), 1 of 5 gelatinous thickenings (20%) and 4 of 24 large plaques (16.7%). For the remaining tissue samples, the mean difference in the proportion of free cholesterol was only 1.5%, and in the proportion of linoleic acid was less than 1%. For the 8 samples which showed significant change, the mean rate of hydrolysis, expressed as micrograms of cholesterol per milligram of defatted dry tissue per hour is shown in

Table III
 SERIES OF SMALL PLAQUES FROM A CASE WITH MYOCARDIAL INFARCTION (male, aged 48)^a

	Cholesterol			P-Lipid (mg/ 100 mg tissue)	Ratio: Free cholesterol Phospholipid	CEFA: % of total			
	(mg/100 mg tissue) ^b		% free						
	Total	Free							
Superficial layer of large fat-filled cells (FFCs); very little extracellular lipid	18.1	4.1	22.7	5.7	0.7	55.0	15.9	2.0	1.1
Large FFCs at all levels; definite extra- cellular lipid in deep layers	20.8	6.2	29.8	6.9	0.9	47.4	20.1	4.4	3.6
Large FFCs at all levels; small area of "amorphous lipid" at center	24.7	7.0	28.6	6.2	1.1	44.3	21.1	4.9	3.7
As in previous lesion; larger area of "amorphous lipid"	30.7	11.2	36.5	11.0	1.0	39.6	24.2	4.0	4.9
Slightly more collagen in cap; large area of "amorphous lipid"	26.9	11.9	44.1	10.8	1.1	37.2	27.4	3.0	4.4

^a Data from Smith *et al.* (1968).

^b Lipid-extracted dry tissue.

Table IV
HYDROLYSIS OF ENDOGENOUS CHOLESTEROL ESTER

Age and sex	Time after death (hr)	Fatty streaks and nodules		Large plaques		Gelatinous thickening	
		Rate of hydrolysis (μ g cholest- terol/mg dry tissue/hr)	% increase of free cholesterol	Rate of hydrolysis (μ g cholest- terol/mg dry tissue/hr)	% increase of free cholesterol	Rate of hydrolysis (μ g cholest- terol/mg dry tissue/hr)	% increase of free cholesterol
59, F	3½	—	—	—	—	0.08	5.5
57, M	4	0.12 ^a 0.44 ^b	9.2 13.5	0.32 —	5.0 —	— —	— —
57, M	4½	0.67	21.8	1.09	8.1	—	—
69, F	13	—	—	8.80	12.3	—	—

^a Scattered fat-filled cells only.

^b Confluent fat-filled cells + amorphous lipid.

Table IV. Significant change in fatty acid pattern occurred in only one sample in which the proportion of linoleate fell by 6.8%; this is the *opposite* direction from the change required to transform the lipid in fat-filled cells into "amorphous" type extracellular lipid. These rates of hydrolysis are lower than the rates calculated from the data of Patelski *et al.* (1967) for pig aorta tissue slices (approximately 20 $\mu\text{g}/\text{mg}$ dry tissue per hour) but mainly higher than the rates calculated from the data of Howard and Portman (1966) for rat and monkey aorta (0.002–0.012 $\mu\text{g}/\text{mg}$ dry tissue per hour). Exogenous substrates were used by both these investigators.

5. Large Plaques Containing Numerous Fat-Filled Cells

The obvious interrelationship between morphology, free cholesterol and cholesterol ester fatty acid pattern, and the failure to demonstrate a hydrolytic mechanism that could account for this, led to an attempt to localize the position in the lesion at which the chemical change occurs. Smith and Slater (1972a) dissected thick cryostat sections into topographically and histologically defined fractions and analyzed the lipids in each fraction. Large plaques were required to obtain sufficient lipid, and it became clear that in human aorta, the typical fatty streak with masses of fat-filled cells and little increase in collagen does not frequently lead to the formation of large atheromas. With only one exception, the "fatty" plaques that were large enough for microdissection had relatively thick collagen caps in which the numerous fat-filled cells were embedded. The lipids in adjacent layers of such plaques, proceeding from the luminal surface downwards, are shown in Table V.

There appears to be no significant change in composition when fat-filled cells start to disintegrate (fractions 1–2) and might be expected to release lysosomal enzymes. At the cap/amorphous lipid junction (fraction 2–3), there is a significant increase in proportion of free cholesterol, but no significant change in cholesterol ester fatty acid composition. Between the upper and middle layers of the amorphous lipid (fractions 3–4) the proportions of free cholesterol increase and of oleic acid decrease—this is the only region of the plaque in which the idea of cholesterol oleate hydrolysis is supported. Between the middle and bottom layers of the amorphous lipid (fractions 4–5) there is a *decrease* in the proportion of free cholesterol and a further significant decrease in the proportion of cholesterol oleate. It is clear from Table V that increase in the proportion of the cholesterol which is free does not necessarily occur in the same region of the plaque as decrease in cholesterol oleate. Furthermore, there is no significant correlation between free

Table V
COMPARISON OF THE LIPIDS IN ADJACENT LAYERS OF LARGE PLAQUES CONTAINING
NUMEROUS FAT-FILLED CELLS IN THE CAP^a

Fraction number	No. of pairs of samples	Cholesterol			Ratio: Free cholesterol Phospholipid	CEFA: % 18:2 in 18:1 + 18:2 fraction
		(mg/100 mg tissue)		% free		
		Total	Free			
1. Intact fat-filled cells in cap	4	38.4	10.0	26.0	1.3	30.1
2. Mixture of intact and disinte- grating fat-filled cells		55.1	21.0	38.1	2.1	33.7
	<i>Difference:</i>	$p = 0.3$	$p = 0.05$	$p = 0.08$	$p = 0.05$	$p = 0.2$
2. Mixture of intact and disin- tegrating fat-filled cells	5	65.2	22.7	34.9	1.9	32.7
3. Disintegrating fat-filled cells mixed with amorphous lipid		138.2	76.6	55.4	2.2	43.1
	<i>Difference:</i>	$p = 0.09$	$p = 0.01$	$p = 0.01$	$p = 0.4$	$p = 0.1$
3. Disintegrating fat-filled cells mixed with amorphous lipid	6	130.7	56.5	43.2	1.4	44.7
4. Upper and middle amorphous lipid free from fat-filled cells		155.9	80.9	51.9	2.2	55.2
	<i>Difference:</i>	$p = 0.6$	$p = 0.03$	$p = 0.03$	$p = 0.1$	$p = 0.015$
4. Upper and middle amorphous lipid free from fat-filled cells	5	172.3	87.0	50.5	2.2	57.9
5. Lower amorphous lipid		83.5	35.3	42.3	1.5	64.1
	<i>Difference:</i>	$p = 0.1$	$p = 0.01$	$p = 0.02$	$p = 0.025$	$p = 0.01$

^a Each layer is compared with that immediately below it in the same plaque.

cholesterol and cholesterol ester fatty acid pattern within any layer of the plaques (Smith and Slater, 1972a); in the cap/amorphous lipid junction area, where maximum hydrolysis might be expected, there is a slight *increase* in oleate and decrease in linoleate, results in line with the tentative findings in the hydrolysis experiments. In the lower amorphous lipid layer, cholesterol linoleate (18:2) accounts for 64% and cholesterol oleate (18:1) for 36% of the 18:1 + 18:2 cholesterol ester fraction; this is similar to the proportion in the cholesterol ester in plasma S_1 0-12 lipoprotein (66% cholesterol linoleate). The proportion of cholesterol oleate increases in each successive layer through the plaque from base to surface, where in the intact fat-filled cells, it accounts for 70% of the 18:1 + 18:2 fraction. In the absence of evidence of preferential cholesterol oleate hydrolysis it seems probable that linoleate-rich cholesterol ester enters directly from the plasma, and "dilutes out" the oleate-rich cholesterol ester derived from the fat-filled cells. In the lowest layer of the amorphous lipid it appears that all the cholesterol ester is derived from plasma; in the top layer where the amorphous lipid just under the cap is mixed with disintegrating fat-filled cells, it can be calculated from the fatty acid distribution that about 60% of the cholesterol ester is derived from the fat-filled cells and 40% directly from plasma.

6. Gelatinous Lesions

These lesions are frequently found in the human aorta, and appear to be early stages in a sequence leading to fibrous plaques. In the smaller thickenings, sections stained for fat typically show only a diffuse sudanophilia or very fine perifibrous lipid droplets on the thick collagen bundles. In larger lesions there may be areas of abundant, coarser perifibrous lipid that most frequently occur in the deep-lying central area where the collagen fibers may lose their linear arrangement and form a network. In the majority of these lesions no fat-filled cells can be detected, and the lipids (Table VI) are similar to the lipids of plasma LD-lipoprotein with a high proportion of linoleic acid (18:2) in the cholesterol esters. Table VI also shows the lipids in three layers of a plaque with a gelatinous cap overlying a central pool of amorphous "atheroma" lipid; no fat-filled cells could be detected in any layer. The proportion of linoleic acid remains extremely high, and there is a striking fall in the proportion of phospholipid in both layers of the amorphous lipid. The proportion of free cholesterol is only slightly increased in the upper layer of the amorphous lipid, but is almost doubled in the lower layer, suggesting that there is either hydrolysis of ester or preferential retention of free cholesterol.

Table VI
COMPOSITION OF THE LIPIDS IN GELATINOUS LESIONS

	Gelatinous thickenings		Layers of a gelatinous plaque		
	Diffuse sudanophilia or weak PFL ^b	Very abundant PFL	Cap	Upper layer of "amor- phous"	Lower layer of "amor- phous"
No. of samples	13	5			
Total cholesterol (mg/100 mg tissue) ^a	4.4 ± 0.4	16.0 ± 5.2	4.9	48.0	79.5
% of cholesterol free	27.1 ± 1.9	20.8 ± 1.7	27.0	29.0	51.6
Percentage of total lipid					
Cholesterol ester	46.7 ± 3.5	63.8 ± 4.7	52.3	60.0	44.5
Free cholesterol	13.0 ± 1.1	12.4 ± 0.1	11.7	14.6	28.4
Phospholipid	29.2 ± 3.5	16.4 ± 5.9	23.5	12.8	11.5
Triglyceride	11.1 ± 1.2	7.5 ± 1.0	12.5	12.6	15.6
CEFA: percentage 18:2 in 18:1 + 18:2 fraction	64.7 ± 1.2	64.9 ± 0.2	69.0	62.5	55.2
Ratio: free cholesterol/phospholipid	0.47 ± 0.04	0.65 ± 0.14	0.51	1.14	2.46

^a Lipid-extracted dry tissue.

^b Perifibrous lipid.

Table VII
"WHITE" FIBROUS PLAQUES CONTAINING NO, OR VERY FEW, FAT-FILLED CELLS

	No. of samples	Cholesterol			Ratio: $\frac{\text{Free cholesterol}}{\text{Phospholipid}}$	CEFA: % 18:2 in 18:1 + 18:2 fraction
		(mg/100 mg tissue)		% free		
		Total	Free			
Fine perifibrous lipid only						
Whole depth of plaque	7	5.8 ± 2.6	1.5 ± 0.8	25.7 ± 3.4	0.5 ± 0.1	62.8 ± 2.2
Plaque in two layers						
Upper	3	3.3 ± 0.6	0.9 ± 0.3	28.2 ± 6.1	0.4 ± 0.3	61.7 ± 2.0
Lower		6.3 ± 1.2	1.5 ± 0.4	24.4 ± 6.8	0.5 ± 0.3	63.0 ± 1.9
Layers through plaque with amorphous lipid center						
Surface of cap	10	2.7 ± 0.2	1.3 ± 0.2	46.9 ± 3.2	0.5 ± 0.1	58.2 ± 3.0
Deep layer of cap	7	8.4 ± 1.6	2.7 ± 0.5	33.6 ± 3.6	0.6 ± 0.1	64.7 ± 3.3
Upper amorphous lipid	10	62.1 ± 7.4	26.1 ± 4.1	42.3 ± 2.6	1.3 ± 0.1	60.8 ± 1.2
Lower amorphous lipid	7	88.4 ± 12.5	47.4 ± 8.6	51.6 ± 2.3	2.1 ± 0.4	58.2 ± 2.3

7. Fibrous Plaques

The composition of the lipid appears to be more closely related to the detailed morphology of the lipid itself than to the overall morphology of the plaque. Thus perifibrous lipid seems to have similar characteristics whether it is located in the loosely structured gelatinous lesions or in the close fibrous tissue of fibrous plaques (Tables VI and VII), whereas the histologically "amorphous" lipid shows marked changes in the proportion of free cholesterol and the ratio of free cholesterol/phospholipid (Table VII). A striking increase in the proportion of sphingomyelin in the phospholipids with increasing severity of atherosclerosis has been

Table VIII
PHOSPHOLIPID PARAMETERS IN TISSUES WITH LIPID OF DEFINED MORPHOLOGY

	Ratio: Free cholesterol Phospholipid	Ratio: Sphingomyelin Lecithin
<i>S_f</i> 0-12 Lipoprotein	0.57	0.9
<i>Tissues with perifibrous lipid</i>		
Normal intima	0.45	0.9
Gelatinous lesions		
Light PFL ^a	0.46	—
Abundant PFL	0.65	—
'White' fibrous lesions		
Whole depth	0.55	0.9
Upper layers	0.43	—
Lower layers	0.54	—
Cap of plaques with underlying amorphous lipid	0.63	1.1
<i>Amorphous lipid</i>		
Fibrous plaques		
Whole amorphous lipid	1.67	5.1
Upper layers	1.27	—
Lower layers	2.05	—
Plaques with numerous fat-filled cells in the cap		
Upper layers	2.00	3.1
Lower layers	1.55	9.0
<i>Lipid within fat-filled cells</i>		
Fatty streaks	0.74	0.9
Fat-filled cells from caps of plaques	1.05	1.0

^a Perifibrous lipid.

reported by many workers (Weinhouse and Hirsch, 1940; Buck and Rossiter, 1951; Steele and Kayden, 1955; Smith, 1960, 1965a; Böttcher and van Gent, 1961); this appears to be particularly associated with the amorphous lipid fraction (Smith *et al.*, 1968). Table VIII summarizes data obtained in this laboratory on the ratios, free cholesterol/total phospholipid and sphingomyelin/lecithin in intimal samples containing lipid of defined morphology. For samples from all sources containing perifibrous lipid, values for free cholesterol/total phospholipid lie between 0.43 and 0.65, and for sphingomyelin/lecithin at about unity; these values are close to the values found for plasma LD-lipoprotein. In samples of amorphous lipid, free cholesterol/total phospholipid increases 3-to 5-fold and sphingomyelin/lecithin increases 3-to 9-fold. Lipid within fat-filled cells shows some increase in the proportion of free cholesterol, but no change in the proportions of sphingomyelin and lecithin.

Thus it appears that amorphous lipid is not merely a massive accumulation of perifibrous lipid, but that it has undergone specific chemical changes. Smith and Slater (1972a) examined the relationship between the percentage of the cholesterol that was free and the cholesterol ester fatty acid composition in successive layers of fibrous plaques (Fig. 3). In the lower layer of the amorphous lipid (Table VII, last line) there is a significant correlation between increase in percentage of the cholesterol that is free and increase in percentage of cholesterol oleate (18:1) in the 18:1 + 18:2 cholesterol ester fraction ($r = 0.901$, $p \ll 0.001$; slope $b = 0.924$, $p < 0.005$). In the upper layer of the amorphous lipid there

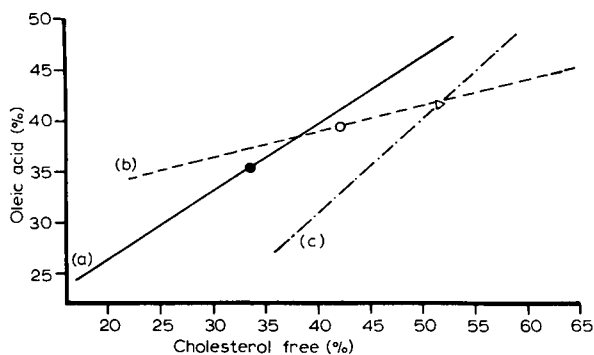


FIG. 3. Regression lines showing the increase in proportion of oleic acid (18:1) in the 18:1 + 18:2 fraction of the cholesterol ester fatty acids with increase in proportion of free cholesterol. Successive layers isolated from large fibrous plaques by microdissection. (a) Lower layers of the fibrous cap. (b) Upper layers of the amorphous lipid (just under the fibrous cap). (c) Deep layers of the amorphous lipid. From Smith and Slater (1972a).

is little variation in the proportion of cholesterol oleate and although the overall range of free cholesterol is large (26–61%), most of the samples (7 of 10) lie within the range 38–48% cholesterol free. In the adjacent deep layers of the fibrous cap there is again a significant correlation between the proportion of cholesterol oleate and the proportion of cholesterol free although the slope is not as steep ($r = 0.738$, $p = 0.02$; $b = 0.668$, $p < 0.01$). The meaning of these linked changes is not clear, but they could be indicative of preferential hydrolysis of cholesterol linoleate; this would be compatible with our preliminary incubation experiments and with the findings of Bowyer *et al.* (1968). However, these workers used exogenous substrates and because of the difficulties of uniform presentation, observed differences in rates of hydrolysis of different cholesterol esters must be interpreted with caution.

III. Concentration of Plasma LD-Lipoprotein in Normal Intima and Lesions

It has been known for many years that plasma constituents are present in the artery wall; conventional immunoelectrophoresis (Grabar and Williams, 1953; Grabar, 1959) of extracts has demonstrated β -lipoprotein and virtually all plasma components (Ott *et al.*, 1958; Gerö *et al.*, 1961; Tracy *et al.*, 1961; Klimov *et al.*, 1974). Investigators have demonstrated LD-lipoprotein by immunofluorescent microscopy in normal intima and lesions of various types (Kayden *et al.*, 1959; Watts, 1963; Kao and Wissler, 1965; Woolf and Pilkington, 1965; Choi, 1966; Knieriem *et al.*, 1967; Walton and Williamson, 1968). In a valuable series of studies More, Haust, and their colleagues have also demonstrated albumin and HD-lipoproteins and compared their distribution with fibrinogen/fibrin in normal intima, fatty streaks, and gelatinous lesions (reviewed by Haust, 1968, 1971). In this laboratory we attempted to obtain *quantitative* measurements of the amount of lipoprotein in extracts of intima. VLD- and LD-lipoprotein fractions were separated by preparative ultracentrifugation and analyzed for cholesterol, and for intact lipoprotein by radial immunodiffusion (Smith and Slater, 1970). Substantial amounts of LD-lipoprotein were found in most tissue samples, but we were unable to demonstrate immunologically intact VLD-lipoprotein although the ultracentrifuge fraction contained a large amount of cholesterol. Unequivocal demonstration of VLDL is difficult because of immunological cross reactions with both LDL and HDL. The low proportion of triglyceride found in both normal intima and lesions suggests that it probably does not enter the intima intact in any quantity. French (1963)

and Schoeffl and French (1968) have published pictures of chylomicra on the endothelium which in some instances appear to be undergoing lysis.

Recently Smith and Slater (1971, 1972b) have developed a method for the measurement of immunologically and electrophoretically intact plasma constituents in very small tissue samples by electrophoresis directly from the minced tissue into an antibody-containing gel. This is a development of the technique of Ressler (1960); the antibody-antigen complex precipitates as a rocket-shaped peak with an area proportional to the concentration of antigen (Clarke and Freeman, 1966). By use of different antisera, aliquots of individual tissue samples have been assayed for LD-lipoprotein and albumin (Smith and Slater, 1972b) and fibrinogen (Smith *et al.*, 1973). The assays can be performed with less than 1 mg defatted dry tissue so that the entire sample of intima can lie within 2 or 3 mm of the histological control sections, and different layers of normal intima, and different layers and regions of lesions can be compared (Smith and Slater, 1973a,b). After electrophoresis the tissue samples are recovered, the residual lipids (not mobile in the electric field) extracted and measured, and the defatted dry weight of the tissue determined. The results obtained using this technique are discussed in the following section.

A. CONCENTRATION OF LD-LIPOPROTEIN IN NORMAL INTIMA

1. Relationship to the Patient's Serum Cholesterol Level

In 30 subjects coming to autopsy in whom there were areas of normal intima in the descending thoracic or upper abdominal regions of the aorta, blood samples were obtained during the week before death, so that a direct comparison could be made between the lipoprotein levels in intima and serum. The samples of intima were assayed together with 10 μ l standard lipoprotein preparation and 10 μ l of the patient's own serum on the same immunoelectrophoresis plate; this allows quantification of intimal lipoprotein in two ways—an absolute measure of lipoprotein concentration in terms of the standard lipoprotein preparation, and quantification in terms of the volume of the patient's own serum or plasma from which the lipoprotein was derived. In 24 subjects, aged 29–69, with no history of hypertension, the absolute amount of lipoprotein in the intima is very highly correlated with the patient's serum cholesterol level during the week before death; the regression is shown in Fig. 4a ($r = 0.971$; $p \ll 0.001$). However, when the lipoprotein is expressed in terms of the volume of the patient's own plasma from which it was derived

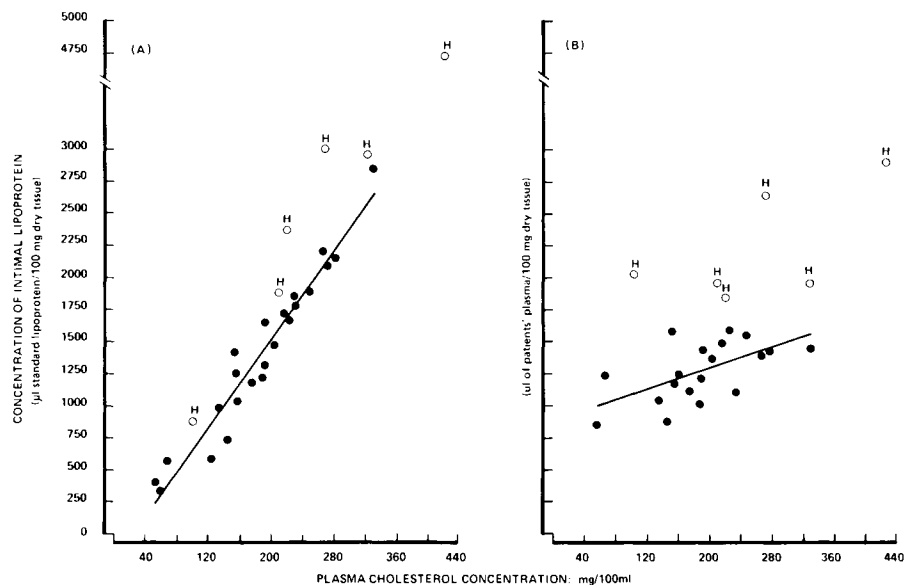


FIG. 4. Relation between the concentration of LD-lipoprotein in normal intima and the subject's serum cholesterol level during the week before death. (A) Expressed as volume of standard lipoprotein preparation. (B) Expressed as the volume of the patient's own plasma from which the lipoprotein has been derived. Hypertensive patients (open circles marked H) are not included in the regression calculations.

(Fig. 4b) the correlation with serum cholesterol level is much lower ($r = 0.589$). In addition to the 24 normotensive subjects, there were six known hypertensives from whom normal intima and blood samples were available; these have not been included in the calculation of the regression lines, but are shown on Figs. 4a and b as open circles marked H. The volume of plasma from which the intima lipoprotein was derived is significantly higher in the hypertensive than in the normotensive group (mean normotensive $1349 \pm 95 \mu\text{l}$ (SEM) patient's plasma /100 mg defatted dry intima; mean hypertensive $2166 \pm 154 \mu\text{l}$ /100 mg; $p < 0.001$).

An average 1 cm^2 sample of "normal" intima with a thickness of $200 \mu\text{m}$ contains lipoprotein derived from about $20 \mu\text{l}$ of the patient's own plasma; thus, very surprisingly, the LD-lipoprotein is at approximately the same concentration in the intima as in the plasma. Smith and Slater (1972b) found that increasingly vigorous washing *increased* the apparent concentration of arterial lipoprotein, so this similarity in concentration does not arise from massive surface contamination.

These findings suggest the concept that lipoprotein from a rather constant amount of plasma is retained in the intima; if the plasma has a low cholesterol level the amount of intimal lipoprotein will be low, whereas if the plasma has a high cholesterol level the amount of intimal lipoprotein will be high. Since hypertension increases the amount of plasma in the intima, the effect of hypertension and hypercholesterolemia on intimal lipoprotein will be additive, and this parallels the findings of the Framingham study on the relationship between serum cholesterol, blood pressure level, and ischemic heart disease (Dawber *et al.*, 1962).

In the lipoprotein assay, the immunoelectrophoresis plates are stained

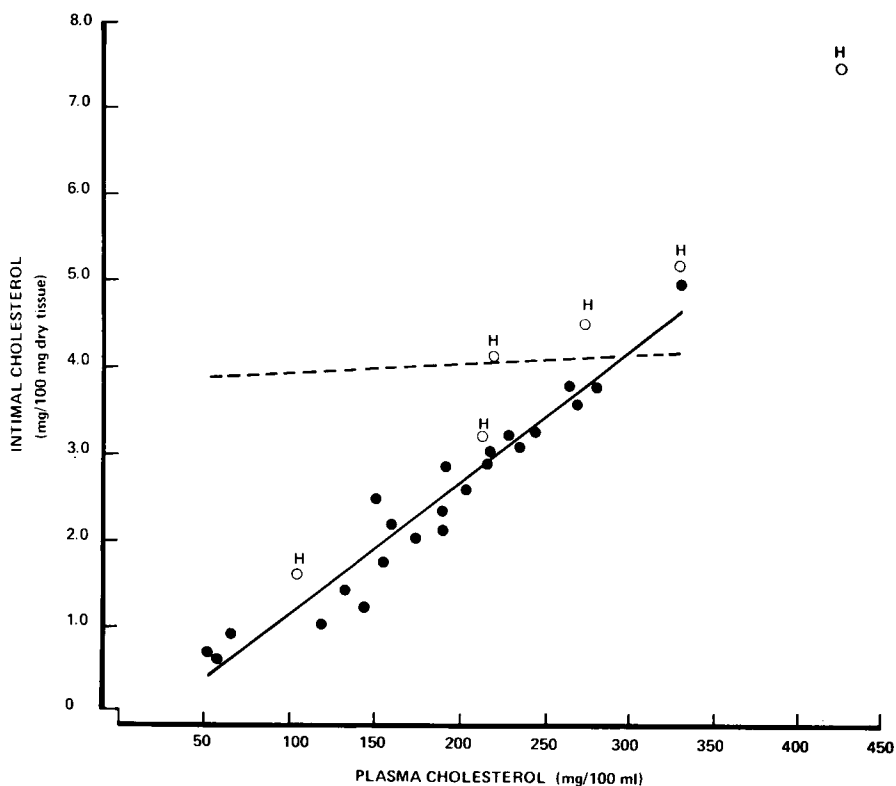


FIG. 5. Relation between the concentration of intact lipoprotein cholesterol (—) in the intima (calculated from the standard lipoprotein) and the patient's serum cholesterol level. Hypertensive patients (open circles marked H) are not included in the regression calculation. The regression line for the residual cholesterol left in the tissue after electrophoresis is also shown (----).

for lipid with oil red O, and the lipoprotein antibody peaks derived from nearly all intimal samples have the same intensity of staining as the plasma lipoprotein standard. In preliminary experiments in which the immunopeaks from tissue samples and the patient's serum were cut out of the agarose and their cholesterol contents compared, the mean concentrations of cholesterol per unit area of immunopeak derived from tissue and the patient's serum were not significantly different. It therefore appears valid to assume that the intimal lipoprotein has the same composition as serum lipoprotein, and on this basis, the concentration of lipoprotein cholesterol in intima has been calculated (Fig. 5). For every 100 mg/100 ml increase in serum cholesterol, the lipoprotein-bound cholesterol in normal intima increases by 1.5 mg/100 mg dry tissue; residual (electrophoretically immobile) tissue cholesterol shows no correlation with plasma cholesterol or intimal lipoprotein levels.

2. Relationship to Age

Because of the high correlation between intimal and plasma lipoprotein levels, the effect of age can only be examined in terms of volumes of the patient's plasma in the intima. There is a small increase of 113 μ l/100 mg defatted dry tissue/decade over the age range 29 to 69 years ($p = 0.03$); this amounts to about 0.1 mg ester cholesterol compared with the total ester cholesterol increase of about 0.7 mg/decade previously found, so that intact LD-lipoprotein accounts for about 10% of the age-related increase in cholesterol ester. The two studies are not, however, strictly comparable because the original cholesterol measurements were taken on intimal samples stripped in a single layer on the first *intact* elastic lamina (Smith *et al.*, 1967), whereas the lipoprotein measurements were made on the upper (luminal) layer of the intima, stripped on the first *fragmentary* elastic lamina (Smith and Slater, 1972b). In these upper intimal samples, the total residual cholesterol increases at a slower rate with a wide scatter (0.55 mg/decade, $p = 0.04$). In 17 samples, analyses are also available on the lower layer, in which the concentration of lipoprotein is only about 16% of the concentration in the upper layer, and which contains the fragmentary elastic laminae. In this layer, residual cholesterol is highly correlated with age ($r = 0.795$), increasing at a rate of 0.99 mg/decade ($p = 0.001$). Thus the age-related increase in total cholesterol occurs mainly in the region of the fragmentary elastic laminae, a result in agreement with many histological observations and other chemical studies (Kramsch and Hollander, 1973; Smith, 1974).

The relationship between serum cholesterol and intimal lipoprotein concentrations (Fig. 4a) seems to be maintained down to age 29, but in the two younger subjects, from whom blood samples were obtained,

the intimal lipoprotein was much lower. In a girl aged 8 years, with serum cholesterol of 94 mg/100 ml, no lipoprotein could be detected in the intima although very large tissue samples were used; no lipoprotein was found in another child of 8 with unknown serum cholesterol. In a boy aged 14, with serum cholesterol of 290 mg/100 ml, the intima contained 150 μ l standard LP per 100 mg tissue, which is only 6% of the expected level; comparable concentrations were found in two boys aged 17 with unknown serum cholesterol levels. Thus it appears that in the first and second decades, there is very little lipoprotein in the intima and this observation agrees with the findings of Kao and Wissler (1965) using immunofluorescent microscopy. It is not clear whether this indicates that the endothelium is not as "leaky," or that a certain minimum thickness of subendothelial connective tissue is required for accumulation of measurable amounts of lipoprotein.

B. DISTRIBUTION OF LD-LIPOPROTEIN IN LESIONS

Because of the close relationship between intimal lipoprotein concentration, blood pressure, and serum cholesterol levels the concentration of lipoprotein in the lesions has little meaning unless it is related to adjacent normal intima or, if distribution within plaques is examined, to other areas of the same plaque.

1. Change in Concentration of LD-Lipoprotein in Lesions

In Table IX the overall concentration of lipoprotein in different types of lesion is presented as a percentage of the concentration in normal intima from the same aorta. In fatty streaks the concentration of lipoprotein is very low, only about a quarter of the normal level; this is a constant finding and confirms previous results on extracts (Smith and Slater, 1970). In intima which is gelatinous, but not thickened beyond normal limits, lipoprotein is increased about 50%; in definite gelatinous thickenings it is doubled and in the periphery of plaques it is increased between 2- and 4-fold. The larger plaques shown in the lower part of the table were analyzed in two layers, and the results have been recombined to give an estimate of the overall lipoprotein concentration. The group of fibrous plaques which have not accumulated lipid still show a marked increase in lipoprotein. Only the fibrous plaques with dense collagenous caps and an underlying pool of amorphous "atheroma" lipid show a fall in lipoprotein concentration. Thus with the exception of the fatty streaks, all lesions which might be regarded as "developing," show increases in lipoprotein concentration.

Table IX
CONCENTRATIONS OF LD-LIPOPROTEIN AND "RESIDUAL CHOLESTEROL" IN LESIONS^a

	No. of sam- ples	Percentage of the normal level	
		Concentra- tion of lipoprotein	Concentration of residual cholesterol
Lesions analyzed in a single layer			
<i>Fatty streaks</i> (containing numerous fat-filled cells)	14	28 ± 5	1319 ± 268
<i>Gelatinous normal</i>	12	142 ± 10	83 ± 12
<i>Gelatinous thickenings</i>	21	196 ± 13	138 ± 19
<i>Gelatinous periphery of plaques</i>	19	363 ± 36	133 ± 24
Reconstituted from lesions analyzed in two layers			
<i>White fibrous plaques</i> (<15 mg/100 mg tissue residual cholesterol)	15	170	95
<i>Plaques with amorphous lipid centers</i>			
Gelatinous	12	221	629
White fibrous	10	55	902

^a Expressed as percentage of the level in normal intima from the same aorta.

2. Distribution of Lipoprotein in Different Layers

The distributions of lipoprotein and residual cholesterol between different layers of normal intima and lesions are summarized in Fig. 6, which shows spatial as well as chemical relationships. In samples divided into two layers, the concentration of LP in the lower layer is expressed as a percentage of the concentration in the upper layer, and shown below each column. The height of the columns represents thickness of the intima in micrometers; the broken line at zero represents the first fragmentary elastic lamina, thus samples below this line contain fragmentary elastic laminae, whereas in samples above the line no elastic tissue could be detected by the Verhoeff stain. In groups 1-4, the upper strip plane was on the first fragmentary elastic lamina and the lower strip plane on the first intact elastic lamina. In groups 5-12, the bottom strip plane was on the first fragmentary elastic lamina and elastin could not be demonstrated by the Verhoeff stain in any of the tissue layers. Groups 1-3 represent normal intima grouped by the thickness of the upper layer; the proportion of lipoprotein in the lower layer remains rather constant at about 16% of the concentration in the upper layer in spite of increasing distance from the lumen. Group 4 samples are of normal thickness but gelatinous; the concentration of lipoprotein in the lower layer is 38% of

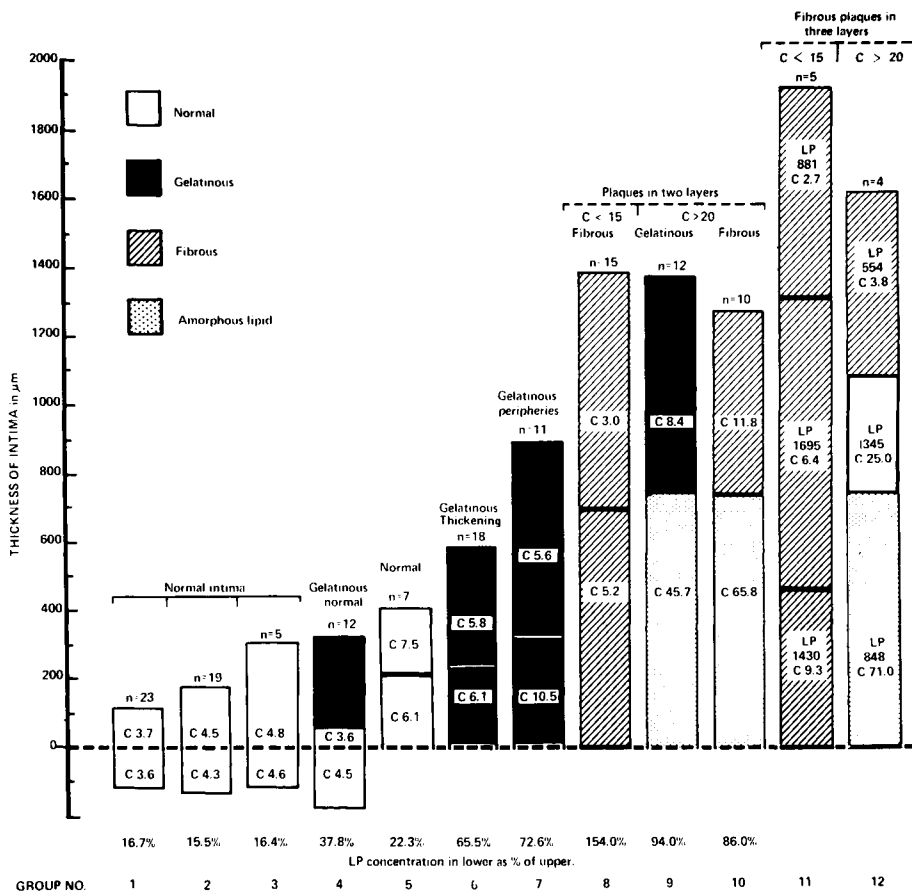


FIG. 6. Distribution of LD-lipoprotein (LP) and residual cholesterol (C) in different layers of normal intima and lesions. The height of the columns represents the thickness of the intima in micrometers, with zero on the first fragmentary elastic lamina (heavy dashed line). Strip planes are represented by horizontal lines. Residual cholesterol in milligrams per 100 mg tissue (C) is shown within each layer. For groups 1-10, the concentration of lipoprotein in the lower layer is expressed as percentage of the concentration in the upper layer, and is shown below each column. For groups 11 and 12 (stripped in 3 layers), the concentration of lipoprotein in microliters per 100 mg tissue (LP) is shown within the columns. Groups 1-3 are normal intimas classified by the thickness of the upper layer: (1) Less than 151 μm ; (2) 151-250 μm ; (3) 251-350 μm . The maximum concentration of residual cholesterol in any layer is used to classify plaques into low-lipid (less than 15 mg/100 mg dry tissue) and high-lipid (more than 20 mg/100 mg dry tissue) groups.

the concentration in the upper layer, thus there is a great increase in permeability.

In group 5, the *lower* strip plane is on the first fragmentary elastic lamina and the upper strip plane is about the middle of the collagenous intima; the intimas are thick with slightly raised residual cholesterol (C) but otherwise appear normal. The lipoprotein concentration in the lower layer is 22% of the concentration in the upper.

In the gelatinous thickenings and gelatinous peripheries of plaques (groups 6 and 7), there is a great increase in the overall concentration of lipoprotein (Table IX) and the proportions in the lower layers rise to 65% and 73%, respectively.

Group 8 are white, fibrous plaques containing only perifibrous lipid and with maximum residual cholesterol less than 15 mg/100 mg. The overall concentration of lipoprotein is slightly greater than normal (Table IX), and the increase occurs mainly in the *lower* layer where the concentration is 154% of the concentration in the upper layer.

Groups 9 and 10 are plaques with amorphous "atheroma" lipid in the deep layers. In group 9 the caps are gelatinous, suggesting that the lesions may be "developing." Overall lipoprotein concentration is high at 220% of normal and is evenly distributed between the gelatinous cap and the "early" amorphous lipid in the lower layer (residual cholesterol 45.7 mg/100 mg). In group 10, dense fibrous caps overlie amorphous atheroma lipid; the overall lipoprotein concentration is only 55% of normal and again it is fairly evenly distributed between the two layers.

Groups 11 and 12 are very thick plaques that have been dissected into three layers; the actual concentrations of lipoprotein are shown within each layer. In both the low and high lipid groups, the maximum concentration of lipoprotein occurs in the middle layer which, in the high lipid group, comprises the junction of the cap with the amorphous lipid; the lowest concentration of lipoprotein is in the surface layers of the fibrous cap.

The concentration of lipoprotein in the upper layer of the media—immediately below the first intact elastic lamina—has been examined in 12 samples. The highest level of 58 μ l standard LP/100 mg (16% of the concentration in the whole intima) was found under a patch of fatty infiltration in which half the intima contained numerous fragmentary elastic laminae. No lipoprotein was detected in media underlying the six samples of normal intima or one of the fatty streaks; in the remaining three fatty streaks the mean concentration was 4% of the concentration in the whole intima. In the media underlying one fatty plaque, the concentration of lipoprotein was 10% of the concentration in the lower layer of the intima and 5% of the concentration in the upper layer. Although

the numbers are very small, these results lend no support to the idea that filtration through the media is a normal route of egress for plasma LD-lipoprotein.

It is clear from the results summarized in Fig. 6 that the pattern of distribution of lipoprotein within the intima is highly complex; presumably the net amount found in any layer is the resultant of differing rates of ingress and egress, of reversible and irreversible binding, and of lipoprotein destruction; however, certain patterns seem to emerge. In normal intima the highest concentration of lipoprotein is in the upper (luminal) half, but in all the gelatinous lesions there is not only a large increase in the concentration of lipoprotein, but a very great increase in its permeation into the deep layers. From the analysis of whole plaques with atheroma centers (Smith and Slater, 1973a), it was thought that low concentrations of lipoprotein were associated with amorphous lipid, but from the last four groups in Fig. 6, it is clear that this is not true. The concentration of lipoprotein is approximately the same in the cap and underlying lipid pool both in gelatinous plaques with a very high lipoprotein content and in fibrous plaques with a low lipoprotein content. In the plaques analyzed, in three layers the distribution of lipoprotein is the same in plaques with low lipid (group 11) and high lipid (group 12). Thus it appears that low levels of lipoprotein are not specifically associated with high concentrations of extracellular lipid. The concentration of lipoprotein appears to be more dependent on topographical position in the lesion and the state of the connective tissue.

IV. Quantitative Relationship Between Plasma Constituents in Intima

The increased concentration of LD-lipoprotein in lesions could result from increased endothelial permeability, increased "binding" of lipoprotein or a combination of both, and studies on the relative and absolute concentrations of other plasma constituents should give some indication of the mechanisms involved. The fluorescent antibody technique is difficult to quantify, but Haust (1968) found albumin in small amounts in the superficial layers only in normal intima, whereas in early lesions it permeated into the deep layers; she was unable to demonstrate fibrin in normal intima, but it was present in all the fatty streaks examined, and in "white, fibrous plaques" where large amounts were demonstrated in the atheromatous "core" which showed only small, scattered deposits with the PTAH stain (Haust *et al.*, 1964). Kao and Wissler (1965) showed that LD-lipoprotein and fibrinogen (or fibrin) paralleled each other in

a large number of tissue samples, but they were unable to demonstrate albumin in the unfixed tissues, and small amounts of HD-lipoprotein were found in only 18 of the 46 samples that contained LD-lipoprotein.

A. LD-LIPOPROTEIN, ALBUMIN, AND FIBRINOGEN IN NORMAL INTIMA AND LESIONS

The quantitative immunoelectrophoretic technique has been used to measure the amounts of LD-lipoprotein, albumin, and fibrinogen in aliquots of the same tissue sample, and they have all been found in substantial amounts in normal intima and lesions. Both albumin and lipoprotein can be assayed against the patient's own serum, giving a direct measure of the "volumes of plasma" from which each is derived. In normal intima, LD-lipoprotein is derived from approximately seven times as much plasma as albumin, which means that their actual concentrations are about the same. Unfortunately samples of the patient's *plasma* are rarely available and the fibrinogen level of most patients is not known. Consequently a ratio is calculated in terms of absolute concentrations of lipoprotein cholesterol and fibrinogen. About 60% of the fibrinogen/fibrin-related antigen (FRA) in the intima can be clotted with thrombin and is presumably intact fibrinogen; the remainder appears to be split products (Smith *et al.*, 1973). The changes in concentration in lesions compared with normal intima from the same aorta are shown in Table X. In fatty streaks containing numerous fat-filled cells, LD-lipoprotein is *reduced* to only 22% of the normal level, but very surprisingly, albumin and FRA also show a small reduction to 60–70% of the normal level. In gelatinous thickenings, LD-lipoprotein and FRA are both increased to more than twice the normal level while albumin shows a rather smaller increase. In the gelatinous periphery of plaques there is a further small increase in albumin and much greater increases

Table X
COMPARISON OF THE CONCENTRATIONS OF LD-LIPOPROTEIN,
ALBUMIN, AND 'FIBRINOGEN' IN LESIONS^a

	Concentration in lesion as percentage of normal		
	Lipoprotein	Albumin	Fibrinogen
Fatty streaks	23	69	62
Gelatinous thickenings	225	177	255
Gelatinous periphery of plaques	352	220	425

^a Expressed as percentage of the concentration in normal intima from the same aorta.

Table XI
RATIOS OF LIPOPROTEIN TO ALBUMIN^a AND OF LIPOPROTEIN CHOLESTEROL TO
FIBRINOGEN IN UPPER AND LOWER LAYERS OF NORMAL INTIMA AND LESIONS

	Lipoprotein/albumin		Lipoprotein cholesterol/ fibrinogen	
	Upper	Lower	Upper	Lower
Normal intima	7.4	4.3	1.9	2.9
Gelatinous thickenings	10.8	8.4	1.5	2.3
Gelatinous periphery of plaques	10.4	18.5	1.9	1.7
White fibrous plaques (residual cholesterol < 15 mg/100 mg tissue)	9.0	16.9	1.3	1.2

^a In terms of volumes of the patient's serum.

in LD-lipoprotein and FRA. The finding that all components increase suggests that in the gelatinous lesions there is a general increase in permeability, but a greater retention of lipoprotein and fibrinogen than of albumin.

The ratios of LD-lipoprotein to albumin and to FRA in the upper and lower layers of normal intima and lesions are summarized in Table XI. In the lower layers of gelatinous lesions and the fibrous plaques with low lipid, in which there is a particularly marked increase in the proportion of LD-lipoprotein there is a 4 fold increase in the ratio of LD-lipoprotein to albumin; only small changes occur in the upper layers. By contrast, the ratio of LD-lipoprotein cholesterol to FRA *decreases*. It could be argued that the results are distorted by fibrinolytic activity with release of "split products" in the deep layers, but Smith *et al.* (1973) found a rather high proportion of thrombin-clottable "fibrinogen" in the lower layers of plaques, and the high level of fibrinogen is compatible with the findings of Haust *et al.* (1964). If there is some mechanism for the specific binding of lipoprotein it appears to act equally upon fibrinogen.

B. RATIOS OF CONSTITUENTS IN PLASMA AND INTIMA AND MECHANISMS OF ENTRY AND RETENTION

There is a remarkable change in the ratios of plasma constituents between the plasma and intimal compartments. The changes in the relationship between albumin, LD-lipoprotein, and fibrinogen are summarized in Table XII. Compared with plasma there is a 7-fold decrease in albumin relative to lipoprotein in normal intima, and a 14-fold decrease

Table XII
PLASMA CONSTITUENTS IN INTIMA^a

	Concentration in plasma (mg/100 ml)	Ratios in plasma and intima		
		Plasma	Intima	
			Normal	Gelatinous
Albumin	4000	14	2	1
LD-lipoprotein	300	1	1	1
Fibrinogen	300	1	$\frac{1}{3}$	$\frac{1}{2}$

^a From Smith (1974).

in gelatinous lesions; fibrinogen shows a 3-fold decrease in normal intima, but is only half the plasma ratio in gelatinous lesions. These changes are the reverse of the changes found in capillary transudates, and highlight the problems of lipoprotein passage through or across the endothelium, and of its retention in the intima.

1. Entry of LD-Lipoprotein into the Intima

Plasma proteins cross the *capillary* endothelium at rates that are inversely proportional to their molecular weights, so that albumin is the largest component of transudates and LD-lipoprotein, with a molecular weight of 2×10^6 , is present in trace amounts only (Dangerfield *et al.*, 1960). Experimental studies on the permeability of endothelia were extensively reviewed by French (1966) and in general, it appears that albumin also crosses arterial endothelium more rapidly than lipoprotein. Our finding (Fig. 4b) that the volume of plasma from which the LD-lipoprotein in the intima is derived remains fairly constant, and that there are constant relationships between lipoprotein and albumin, and lipoprotein and fibrinogen suggests that "whole plasma" may enter the intima and that the change in the ratios of plasma components in intima results from physical interactions affecting their rates of egress. This would imply either that the endothelium is "totally leaky" or that whole plasma is engulfed in pinocytotic vesicles. Recent studies lend support to this latter idea; there appear to be much tighter cell junctions in arterial than in capillary endothelium (Hess and Staubli, 1968; Schwartz and Benditt, 1972; Hess, 1973) and proteins apparently cross the endothelium in pinocytotic vesicles. Stein and Stein (1973) found that horseradish peroxidase (M.W. 40,000; diameter 40 Å) appeared to cross the endothelium entirely in vesicles when at low concentration, but at high concentration, some peroxidase appeared within the cell junctions, an

observation in agreement with Florey and Sheppard (1970). Schwartz and Benditt (1972) also found horseradish peroxidase in vesicles, some of which appeared to open into the cell junctions. Ferritin (diameter 110 Å) and lactoperoxidase (M.W. 82,000) do not appear to enter the junctions, and cross the endothelial cells in vesicles (Florey and Sheppard, 1970; Stein and Stein, 1973). By analogy, Stein and Stein suggest that HD-lipoprotein (diameter approximately 100 Å) and LD-lipoprotein (diameter 200 Å) will also cross *intact* arterial endothelium in vesicles. In perfusion experiments with protein-labeled VLD-, LD-, and HD-lipoproteins they found very little penetration of the endothelium by VLD-, whereas both LD- and HD-lipoprotein appeared in and under the endothelium, but penetrated only the most superficial layers of the media. By contrast, apo-HDL (M.W. about 16,000) was uniformly distributed throughout the media. Schwartz and Benditt (1972) found that horseradish peroxidase had penetrated all layers of the vessel wall within 20 minutes. Adams *et al.* (1968) found labeled albumin, γ -globulin, and cholesterol in all layers of the rabbit aorta 1–4 days after injection. Thus all these different approaches give remarkably concordant results; if whole plasma, with the possible exception of VLDL, is transported in vesicles and the smaller proteins then pass on through the media, the high ratio of lipoprotein to albumin could be explained. The rapid transport of lipoprotein across the endothelium of these normal experimental animals suggest that the failure to detect measurable amounts of LD-lipoprotein in young children springs from the absence of a subendothelial connective tissue layer in which it can accumulate. There is, however, one aspect of the problem about which we have no information, and that concerns the condition of the endothelium in the “normal” middle-aged human artery. Areas of increased permeability and cell turnover have been demonstrated in the aortas of rats and rabbits perfused with dye *in situ* (Björkerud and Bondjers, 1972) and in the aortas of young normal pigs, administered dye *in vivo* (Bell *et al.*, 1972; Caplan and Schwartz, 1973) and it is possible that the endothelium overlying diffuse intimal thickening and gelatinous lesions is “totally leaky.” Since Caplan and Schwartz find different results of dye-uptake experiments *in vivo* and in excised aortas it is difficult to see how the problem can be investigated in human vessels.

2. Binding of LD-Lipoprotein in the Intima

Mucopolysaccharides or glycosaminoglycans (GAG) have been implicated in atherogenesis for many years, but more recent *quantitative* studies indicate a decrease in concentration with increasing severity of atherosclerosis (reviewed by Smith, 1974) and their precise role, if

any, is very difficult to interpret. GAG *in vitro*, including preparations from arterial intima, form complexes with LD-lipoproteins at concentrations that are comparable to the concentration in intima (about 3 mg/cm³) (Amenta and Waters, 1960; Gerö *et al.*, 1961; Bihari-Varga and Végh, 1967; Srinivasan *et al.*, 1970; Iverius, 1972). Thus it seems reasonable to postulate that such complexes will be formed in the arterial wall; Bihari-Varga *et al.* (1968) using thermal decomposition, find evidence of complexes in whole intimal tissue, and Srinivasan *et al.* (1972) report extraction of complexes from "fatty streaks" that have not been defined morphologically. However, Iverius (1972) considers that only dermatan sulfate (chondroitin sulfate B) will form stable complexes at physiological pH and ionic strength. From the scanty and discordant information available on distribution of different GAG in arterial wall (see Smith, 1974) it appears that this is a very minor component, and decreases in lesions.

Data from this laboratory on concentrations of LD-lipoprotein, total GAG, and "deposited" cholesterol in fatty streaks and gelatinous thickenings are summarized in Table XIII; lipoprotein and GAG were not determined on the same tissue samples, but the samples are believed to be comparable. In fatty streaks, LD-lipoprotein falls to about a quarter of the normal level, GAG increases by about 16%, and deposited cholesterol increases about 13-fold and is mainly within fat-filled cells. It is difficult to believe that a 16% increase in GAG could increase *cellular* uptake of cholesterol more than 10-fold, and it is also difficult to understand what role GAG could be playing in the massive reduction in LD-lipoprotein found both by electrophoresis directly from the tissue into an antibody-containing gel, and by saline extraction and radial diffusion (Smith and Slater, 1970, 1972b). It has been reported that GAG-lipoprotein complexes have a high electrophoretic mobility at pH 8.6 and that the lipoprotein retains its antigenic properties (Gerö *et al.*, 1961;

Table XIII
CHANGES IN THE CONCENTRATIONS OF INTACT LD-LIPOPROTEIN,
GLYCOSAMINOGLYCANS (GAG), AND "DEPOSITED"
CHOLESTEROL IN EARLY LESIONS

	Percentage of the normal level	
	Fatty streaks	Gelatinous thickenings
LD-lipoprotein	28% (14) ^a	196% (17)
Total glycosaminoglycan	116% (19)	91% (7)
Deposited cholesterol	1319% (14)	138% (17)

^a Number of samples.

Srinivasan *et al.*, 1970); therefore, even if a complex is formed which does not dissociate at pH 8.6, it should be measurable in the immuno-electrophoretic system. There is some tentative support for the suggestion that lipoprotein may be destroyed at the surface of cells that are taking up cholesterol (Robertson, 1968); this could account for the low level of intact lipoprotein found in intima containing fat-filled cells, and the slight increase in GAG could be secondary to their increased metabolic activity.

In the gelatinous lesions the 40% increase in "deposited" cholesterol is mainly in the form of fine perifibrous lipid droplets and diffuse sudanophilia. The concentration of LD-lipoprotein is doubled, but the concentration of GAG is *decreased* by 10%, making it difficult to postulate that the increase in lipoprotein results from increased complex formation; furthermore, fibrinogen increases to the same extent as lipoprotein (Table XI). Lindner (1972) also reports a decrease in GAG in edema lesions, but finds an increase in the glycoprotein fraction. It is clear that there is no simple relationship between intimal lipoprotein, GAG, and deposited cholesterol, but it is not clear that an hypothesis of specific binding of lipoprotein is required. If albumin is able to pass through the media with minimal hindrance, whereas LD-lipoprotein is virtually unable to enter the media, an area of leaky endothelium allowing increased inflow of whole plasma might be expected to produce a greater increase in lipoprotein than in albumin. *Reduced* levels of GAG, which all investigators find in advanced fibrous plaques (Smith, 1974) might allow more rapid penetration of lipoprotein into the deep layers of the intima.

V. Relationship Between LD-Lipoprotein and the Lipid "Deposited" in Intima

A. COMPARISON OF LIPOPROTEIN AND INTIMAL LIPIDS

1. *Extracellular Lipid in Normal Intima*

In normal intima of young children, phospholipid is the major component and most cholesterol is in the free (unesterified) form; the predominant fatty acids in the small cholesterol ester fraction are palmitic and oleic. By the eighth decade, diffusely thickened but otherwise lesion-free intima, with lipid only in the perifibrous form, contains four times as much lipid of a composition very similar to the S_r 0-12 lipoprotein of plasma (Table I). Because of its composition we have assumed that the additional lipid is derived directly from plasma LD-lipoprotein

(Smith *et al.*, 1967). Calculations of the resultant lipid composition, if varying amounts of LD-lipoprotein are added to the lipids of a child's intima broadly support this idea, although the calculated mixture contains more free cholesterol and phospholipid than is actually found.

The most rapid accumulation of cholesterol with age occurs in the lower layers of the intima, where it increases at a rate of 0.1 mg/100 mg dry tissue per annum. In this layer the concentration of intact lipoprotein cholesterol is about 0.5 mg, so this would represent "deposition" or "irreversible binding" of about 20% per annum of the available lipoprotein pool; in the upper, collagenous layer of the intima, cholesterol accumulates at only half the rate, representing 1–2% of the larger lipoprotein pool. There seems to be no correlation between the concentrations of intact LD-lipoprotein and of residual (electrophoretically immobile) cholesterol in normal intima, suggesting that this "deposited" cholesterol is no longer able to equilibrate with plasma cholesterol. This idea seems to be at variance with the apparently rapid equilibration between plasma and intimal cholesterol oleate and linoleate during the first two decades (Fig. 2). However, the "constant concentration" cholesterol ester of the first two decades (Fig. 1) may be in a different pool (Wilson, 1970)—possibly the cell membrane—from the cholesterol ester that accumulates as fine extracellular droplets from the third decade upwards. It is in the third decade that the adult level of lipoprotein in the intima is reached.

2. Intracellular Lipid in Fatty Streaks

In all aortic fat-filled cells (Table II) and the fat-filled cells of adrenal cortex and skin xanthomas (Dailey *et al.*, 1960; Jepson *et al.*, 1965; Baes *et al.*, 1968; Smith, 1971; Smith and Slater, 1973b; Fletcher, 1973) the cholesterol esters have a highly characteristic fatty acid pattern in which there is a very high proportion of oleic acid and low proportion of linoleic acid, thus differing radically from plasma cholesterol esters. This seems to be the characteristic pattern in which cells esterify cholesterol (Blomstrand *et al.*, 1964) compared with the high linoleate plasma pattern arising from lecithin-cholesterol acyl transferase (LCAT) activity (Glomset, 1962, 1968). Thus the cholesterol ester in fat-filled cells could only be derived from plasma cholesterol ester by very elaborate differential hydrolysis, and there is no conclusive experimental evidence in support of this idea. There is, however, much evidence that arterial intima, including isolated foam cells, synthesize fatty acids (reviewed by Whereat, 1971) and esterify them with cholesterol (Lofland *et al.*, 1965; Day and Wilkinson, 1967; St. Clair *et al.*, 1968, 1969, 1970; Day and Wahlqvist, 1968; Day and Tume, 1970). Oleic acid seems to be a major product of synthesis and of incorporation into cholesterol ester (St. Clair

et al., 1969, 1970). By contrast, cholesterol is synthesized only to a small, and probably not significant extent in older and atherosclerotic arteries (reviewed by Dayton and Hashimoto, 1970) and there is much evidence, some of which will be presently discussed, that it is mainly derived from plasma. The finding that there is a consistently low level of intact LD-lipoprotein in the vicinity of fat-filled cells therefore suggests that they break down the lipoprotein and from it derive cholesterol which is then esterified in the cellular pattern. Fatty streaks containing confluent fat-filled cells accumulate about 20 mg cholesterol/ 100 mg tissue; they would thus need to "process" the lipoprotein from 10–15 ml normo-cholesterolemic plasma if all the cholesterol in the lipoprotein were used. If one assumes that a human fatty streak takes 6 months to develop, the cells would have to utilize lipoprotein from 100 μ l plasma per day, which is less than 10% of the normal intimal lipoprotein level, and would be unlikely to produce the 70% decrease observed. This implies either that the cells are accumulating cholesterol much more rapidly, or that only part of the lipoprotein cholesterol is utilized. We do not know if there is hydrolysis and reesterification of the ester fraction, or if only the free cholesterol of the lipoprotein is utilized. If only free cholesterol were utilized there should be a disproportionately low free cholesterol and high ester cholesterol in the tissue surrounding the fat-filled cells. Lang and Insull (1970) fractionated fatty streaks into lipid droplets and tissue residue; in the residue there were normal proportions of free and ester cholesterol, and a normal free cholesterol/phospholipid ratio. This provides some indirect evidence that both free and ester cholesterol are utilized. Peters *et al.* (1973) have found "light lysosomes" containing large amounts of cholesterol in cholesterol-fed rabbits, but these seemed to be associated with rather low levels of esterase activity. Portman and Alexander (1972) have described "light microsomes" which increased early during the development of atherosclerosis and had a high ratio of free cholesterol to phospholipid, and high proportion of sphingomyelin in the phospholipid.

3. Extracellular Lipid in Fibrous Plaques

In the gelatinous and fibrous lesions containing only perifibrous lipid (Tables VI and VII) all the parameters of lipid composition are very similar to LD-lipoprotein; the concentration of lipoprotein is approximately double the concentration in normal intima (Table IX), and intact lipoprotein can account for half the total cholesterol in the lesion. In plaques with a gelatinous cap overlying an amorphous "atheroma" lipid core, intact lipoprotein accounts for about half the total cholesterol in the cap, but only 12% of the cholesterol in the amorphous lipid, and the

proportions are lower in the "white" fibrous plaques (Fig. 6). From plaques divided into three layers (Fig. 6, groups 11 and 12) it is clear that there is considerable variation in lipoprotein concentration between the layers with the maximum level in the mid-zone. However, variation in the relative proportions of the lipid present, as intact lipoprotein and as deposited residual lipid, cannot account for the large differences in the percentage of the cholesterol that is free, or in the changing ratio of free cholesterol/phospholipid. Disappointingly, these analyses seem to give no clue to the mechanism of lipid deposition from the lipoprotein.

4. Relationship Between Cholesterol and Phospholipid

An intriguing feature of the lipid accumulating in fibrous lesions is a progressive increase in the ratio of free (or total) cholesterol to phospholipid. Since it appears that the cholesterol is derived directly from LD-lipoprotein, any increase above the mass ratio of 0.57 found in the lipoprotein, must indicate that lipoprotein phospholipid is being removed.

In normal intima, free cholesterol/phospholipid remains remarkably constant over the first six decades at a mass ratio of 0.4–0.5, which is close to a 1:1 molar ratio. It is only in the eighth decade, where total cholesterol increases above 8 mg/100 mg tissue, that the ratio reaches or slightly exceeds the ratio in LD-lipoprotein.

Fleischer and Rouser (1965) report 1:1 as the molar ratio in plasma membrane. Regardless of the morphology of the surrounding tissue, this ratio seems to be maintained in all tissue samples in which total cholesterol does not exceed approximately 8 mg/100 mg tissue and the lipid, when demonstrated by light microscopy, is in the form of fine, perifibrous droplets (Table VIII). The ratio of cells to connective tissue elements differs greatly between normal intima with 25% collagen, gelatinous lesions with 30% collagen, and fibrous plaques with more than 40% collagen (Smith, 1965b, 1974), so this apparent maintenance of the plasma membrane ratio may be fortuitous and instead reflect a physicochemical relationship (Small, 1970). In samples in which the lipid is morphologically extracellular the ratio increases as soon as the cholesterol rises above about 8 mg/100 mg tissue—increases of up to 3-fold where there is no marked change in the proportion of the cholesterol which is free, and of up to 5-fold where there is a large change in free cholesterol. Thus it appears that at least two-thirds of the incoming phospholipid in the lipoprotein must be eliminated, and since there is active phospholipid synthesis in the arterial wall the proportion eliminated may be even higher. Phospholipid metabolism in the arterial wall was extensively reviewed and discussed by Portman in 1970 and will not be recapitulated here.

It may, however, be of interest to examine the relationship between the ratios of free cholesterol/total phospholipid and of sphingomyelin/lecithin in Table VIII. In all samples with "normal" amounts of periferous lipid, the ratio of free cholesterol/phospholipid is about 0.5 and sphingomyelin/lecithin about 1.0; the sphingomyelin/lecithin ratio is about 1.0 in fat-filled cells. It is only in the amorphous lipid, where the very large amount of cholesterol ester appears to be derived directly from LD-lipoprotein, that there are substantial increases in both ratios. This would be compatible with extensive hydrolysis of lecithin (Eisenberg *et al.*, 1969) and suggests that the large amount of sphingomyelin that accumulates in amorphous lipid is derived mainly from LD-lipoprotein.

5. Accumulation of Plasma-Type Lipid in Other Tissues

The accumulation with age, of lipid showing characteristics of LD-lipoprotein composition, has been observed in a number of connective tissues (Andrews, 1962; Broekhuysse and Kuhlmann, 1972; Broekhuysse, 1972; Crouse *et al.*, 1972; Smith and Slater, 1973b). Crouse *et al.* (1972) showed large increases in cholesterol in biceps and psoas tendons over the age range 20–80 years, with a mean of about 1 mg/100 mg dry tissue in the 60–80 age group; the largest increase was in esterified cholesterol, which increased to between a half and a third of the total cholesterol. More detailed studies have been reported on the sclera and cornea of the eye; over the age range 3–68 years, Broekhuysse (1972) found slow, linear increases in free cholesterol and total phospholipid and a much more rapid increase in cholesterol ester, giving curves very similar to aging aortic intima; there was a progressive increase in the proportion of sphingomyelin.

In this laboratory we have compared lipids in other connective tissues with artery wall; in Achilles tendon and ligamentum flavum, lipid can be seen microscopically as very fine sudanophilic droplets lying between the bundles of collagen or elastin—the elastic ligament appears to have no greater affinity for lipid than the collagenous tendons. In the sclera of the eye, lipid appears as rather coarser droplets lying between the connective tissue fibers; in this tissue, cholesterol ester concentration is most precisely correlated with age ($r = 0.99$), increasing from 0.9 mg/100 mg dry tissue at age 51 to 3.1 mg at age 83 (Smith and Slater, 1973b). The lipids in sclera and cornea are very similar both in concentration and composition, and the lipid patterns in all the connective tissues examined are comparable to aortic intima: 30–35% of the cholesterol free, 55–60% cholesterol linoleate in the oleate + linoleate fraction, and a sphingomyelin/lecithin ratio close to unity. The only consistent difference

is a higher free cholesterol/phospholipid ratio that may reflect the small number of cells in these tissues. Thus it is clear that the age-related accumulation of perifibrous lipid droplets, which appears to be derived from LD-lipoprotein, is not confined to the artery wall.

B. LIPOPROTEIN CONCENTRATION AND DYNAMIC STUDIES ON INFLUX AND EFFLUX

1. *Exchange of Free Cholesterol Between Lipoproteins and Membranes*

Since the measured concentration of plasma LD-lipoprotein in intima must be the resultant of influx, efflux, reversible binding, and irreversible binding or destruction, it tells us nothing about the rate of entry. A high concentration could arise from increased endothelial permeability, an increase in "reversible binding" producing a decreased rate of efflux, or reduction in the rate of lipoprotein destruction. There have been numerous studies on the uptake of isotopically labeled lipoprotein by arterial wall in attempts to disentangle these factors.

The majority of investigators have used isotopically labeled cholesterol that has been incorporated into the lipoprotein either *in vitro* or *in vivo*. However, it seems probable that interpretation of results obtained with this system are complicated by the rapid exchange of the free cholesterol in the lipoprotein with free cholesterol in the tissue. This exchange was first demonstrated with red cell membrane (Hagerman and Gould, 1951; Eckles *et al.*, 1955; Gould *et al.*, 1955¹) and is now known to occur with a large range of other membranes including mitochondria, intestinal microvilli, liver plasma membrane, muscle sarcolemma, and microsomes (Graham and Green, 1967); it also occurs between different lipoprotein fractions (Roheim *et al.*, 1963). The problem of quantitative interpretation of apparent uptake in the presence of this exchange were discussed by Gould *et al.* (1963). In 1966, papers indicating that the passage of labeled cholesterol into the artery wall was not an energy-requiring process were published by both Hashimoto and Dayton and by Newman and Zilversmit. Both groups found that the entry of free cholesterol was much greater, relative to its concentration in plasma, than the entry of esterified cholesterol, and both concluded that at least part of the free cholesterol might be transferred by physicochemical exchange. Newman and Zilversmit (1962) published data from an experiment in which rabbits were fed a high cholesterol diet containing cholesterol-4-¹⁴C so that uptake of plasma cholesterol could be observed

¹ See also Ashworth and Green (1964).

during atherogenesis. From their results, the increment of cholesterol in the aorta can be calculated, and the expected number of counts compared with the observed counts. At 21 days, the counts in the free cholesterol fraction were greater than the counts predicted from the observed cholesterol increment, suggesting that between 25% and 40% of the free cholesterol represented exchange with preexisting tissue cholesterol.

2. Measurement of Isotope in Human Arteries

Several investigators have administered tracer doses of labeled cholesterol to terminal patients and measured the isotope in the artery wall at autopsy (Rabinowitz *et al.*, 1960; Field *et al.*, 1960; Chobanian and Hollander, 1960; Gould *et al.*, 1963); Scott and Hurley (1970) administered LD-lipoprotein labeled with radioactive iodine in the peptide, so that problems of cholesterol exchange were avoided. In "normal appearing" intima, Field *et al.* (1960) found no cholesterol- ^{14}C at 2.5 days after feeding, traces only at 18 days, and substantial amounts from 27 days upwards. From the number of counts in the wall and the counts per 100 ml of plasma (unfortunately individual plasma and intimal cholesterol values were not given), a crude estimate can be made of the "volume of plasma" in the intima. Although the plasma specific activity ranged from 268 to 67 counts/minute/mg cholesterol, the three subjects examined at 27, 38, and 88 days gave quite similar results of $1000 \pm 300 \mu\text{l}/100 \text{ mg dry intima}$ when the calculation is based on ester cholesterol, and $1700 \pm 200 \mu\text{l}$ when based on free cholesterol. The figure from cholesterol ester is entirely compatible with the "volumes of plasma" that we have found by direct measurement of lipoprotein (page 24), and suggests that there is complete equilibration between plasma and intimal lipoprotein in 3 to 4 weeks. The difference between the estimates based on ester and free cholesterol could be taken to indicate that about 40% of the free cholesterol counts represent exchange with tissue cholesterol, which is close to the proportion estimated from the experimental data of Newman and Zilversmit (1962).

Scott and Hurley (1970) presented their data as a ratio of counts per gram of tissue to counts per milliliter of plasma; from the results of Smith and Slater (1972b, 1973a) it appears that the concentration of lipoprotein in the upper layers of the intima is about equal to the concentration in plasma, but for "intima plus inner media" preparations the concentration would only be 30–40% of the plasma concentration, thus a ratio of 0.3–0.4 should indicate equilibrium. In Scott and Hurley's published curves, the ratio rises rapidly and by day 15—their longest time interval—the ratio had reached 0.3, suggested equilibration in 2 to 3 weeks. Thus the results of these two human studies are in close agreement, and compatible with our concentration data.

Unfortunately, there seems to be no adequate data on lesions; Field *et al.* (1960) found about three times as many counts for both free and ester cholesterol in "plaques" as in normal intima, whereas Rabinowitz *et al.* (1960) found fewer counts in plaques; results expressed as specific activity are confusing because of dilution by preformed deposits of cholesterol.

3. *Experimental Studies on Cholesterol Influx and Efflux*

Hashimoto and Dayton (1966) found a much greater influx of free cholesterol than of ester cholesterol into normal rat aorta *in vitro*; their results can be used to calculate the "volume of plasma" in the whole aorta and give figures of 16 μ l when based on ester cholesterol and 140 μ l when based on free cholesterol. Dayton and Hashimoto's (1966) results on uptake *in vivo* are again 16 μ l when based on ester cholesterol, but only 60 μ l when based on free cholesterol, which is a ratio of 3.7. It seems possible that the exchange reaction is accelerated, or the uptake of whole lipoprotein reduced under *in vitro* conditions, but this idea is not supported by results from other laboratories. Thus data from Newman and Zilversmit (1966) give ratios of 2.4 and 3 after oral administration of labeled cholesterol 2 days before sacrifice, compared with 1.6 and 3.3 after *in vitro* incubation with labeled plasma, and the results of Day *et al.* (1970) give ratios of 1.8 both after feeding and *in vitro*; both these groups used atherosclerotic rabbits.

There is now much experimental evidence indicating that severity of atherosclerosis greatly influences influx of free and ester cholesterol; unfortunately, valuable information has been lost through the failure of investigators to define the morphology of the lesions with which they work. Schwenk and Stevens (1960) fed a moderate cholesterol diet to rabbits for 12 weeks, substituting feeding with cholesterol- ^{14}C for one week at different times during the experiment. Radioactivity incorporated during the first week had virtually disappeared by week 12, but radioactivity incorporated during week 7 showed no decrease over the next 5 weeks. Newman and Zilversmit (1962) found increases in both influx and net accumulation with increasing time of cholesterol feeding, and Bell *et al.* (1970) found that rates of both influx and efflux were proportional to the aortic cholesterol concentration. Lofland and Clarkson (1970) fed atherosclerotic pigeons labeled cholesterol for periods up to 30 days, sacrificing them at intervals between 18 and 122 days. The aortas were divided into normal areas, "fatty streaks," and plaques; counts per gram of wet tissue increased about 4-fold between normal intima and plaques for free cholesterol, and 25-fold for ester cholesterol; the ratio, free/ester counts fell from 4.8 in normal intima to 0.7 in plaques. The influx pattern in the plaques would thus be compatible with the idea of

greatly increased permeability to whole lipoprotein. Unfortunately, inasmuch as the pigeon "fatty streaks" seem to differ greatly from human fatty streaks, increase in lipid concentration is relatively small and 70% of the cholesterol is free, they shed no light on one of the most striking characteristics of the human lesion—the low concentration of intact lipoprotein.

VI. Conclusions

In atherosclerosis research there are almost as many methods of approach as there are research workers. The experimental systems have ranged from man through a host of experimental animals to tissue cultures. Results are expressed in every conceivable way. Superficially, the field appears to be hopelessly confused and results frequently contradictory. However, after this attempt to effect some sort of synthesis between the lipids accumulating in arterial tissue and the plasma lipoproteins from which they, at least in part, are derived, the author is left with the feeling that the available data are more often compatible than incompatible.

The accumulation of cholesterol ester with the characteristics of plasma cholesterol ester in aging normal intima and in plaques, is compatible with the demonstration of plasma lipoproteins in artery wall by immunofluorescent microscopy. This has led in turn, to the quantitative measurement of intimal lipoprotein, and the concentration estimated by immunoelectrophoresis is remarkably corroborated by the uptake of radioactive cholesterol ester. The immunoelectrophoretic assay shows that the concentration of lipoprotein in normal intima is highly correlated with plasma lipoprotein level, and radioisotope studies suggest complete equilibration between plasma and intimal lipoprotein in two to three weeks. In "early" fibrous lesions the amount of intact lipoprotein is greatly increased providing, particularly in hypercholesterolemic subjects, a substantial reservoir of lipoprotein from which lipid could be "deposited."

There is, however, no information on the nature of this "deposition." It is not even clear if the apoprotein is still attached to the perifibrous and amorphous lipid droplets or if it is the removal or destruction of the apoprotein that causes the lipid to precipitate. It is in this area that information is now particularly needed to explain the accumulation of pathological amounts of lipids derived from plasma lipoproteins in arterial tissue.

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References

- Abdulla, Y. H., Adams, C. W. M., and Bayliss, O. B. (1969). *J. Atheroscler. Res.* **10**, 149.
- Adams, C. W. M. (1969). *Advan. Lipid Res.* **7**, 1.
- Adams, C. W. M., Virág, S., Morgan, R. S., and Orton, C. C. (1968). *J. Atheroscler. Res.* **8**, 679.
- Amenta, J. S., and Waters, L. L. (1960). *Yale J. Biol. Med.* **33**, 112.
- Andrews, J. S. (1962). *Arch. Ophthalmol.* **68**, 264.
- Ashworth, L. A. E., and Green, C. (1964). *Biochim. Biophys. Acta* **84**, 182.
- Baes, H., van Gent, C. H., and Pries, C. (1968). *J. Invest. Dermatol.* **51**, 286.
- Bell, F. P., Lofland, H. B., and Stokes, N. A. (1970). *Atherosclerosis* **11**, 235.
- Bell, F. P., Somer, J. B., Craig, I. H., and Schwartz, C. J. (1972). *Atherosclerosis* **16**, 369.
- Bihari-Varga, M., and Végh, M. (1967). *Biochim. Biophys. Acta* **144**, 202.
- Bihari-Varga, M., Simon, J., and Gerö, S. (1968). *Acta Biochim. Biophys.* **3**, 375.
- Björkerud, S., and Bondjers, G. (1972). *Atherosclerosis* **15**, 285.
- Blomstrand, R., Gürtler, J., and Werner, B. (1964). *Acta Chem. Scand.* **18**, 1018.
- Böttcher, C. J. F. (1961). In "Drugs Affecting Lipid Metabolism" (S. Garattini and R. Paoletti, eds.), p. 54, Elsevier, Amsterdam.
- Böttcher, C. J. F., and van Gent, C. M. (1961). *J. Atheroscler. Res.* **1**, 36.
- Böttcher, C. J. F., Boelsma-van Houte, E., ter Haar Romeny-Wachter, C. C., Woodford, P. F., and van Gent, C. M. (1960). *Lancet* **ii**, 1162.
- Bowyer, D. E., Howard, A. N., Gresham, G. A., Bates, D., and Palmer, B. V. (1968). *Progr. Biochem. Pharmacol.* **4**, 305.
- Broekhuysse, R. M. (1972). *Biochim. Biophys. Acta* **280**, 637.
- Broekhuysse, R. M., and Kuhlmann, E. D. (1972). *Exp. Eye Res.* **14**, 11.
- Buck, R. C., and Rossiter, R. J. (1951). *Arch. Pathol.* **51**, 224.
- Caplan, B. A., and Schwartz, C. J. (1973). *Atherosclerosis* **17**, 401.
- Chobanian, A. V., and Hollander, W. (1960). *Clin. Res.* **8**, 179. (Abstr.)
- Choi, J. H. (1966). *Fed. Proc., Fed. Amer. Soc. Exp. Biol.* **25**, 665. (Abstr.)
- Clarke, H. G. M., and Freeman, T. (1966). *Protides Biol. Fluids, Proc. Colloq.* **14**, 503.
- Clarkson, T. B. (1963). *Advan. Lipid Res.* **1**, 211.
- Crouse, J. R., Grundy, S. M., and Ahrens, E. H. (1972). *J. Clin. Invest.* **51**, 1292.
- Dailey, R. C., Swell, L., Field, H., and Treadwell, C. R. (1960). *Proc. Soc. Exp. Biol. Med.* **105**, 4.
- Dangerfield, W. G., Smith, E. B., Kinmonth, J. B., and Taylor, G. W. (1960). *J. Clin. Pathol.* **13**, 76.
- Dawber, T. R., Kannel, W. B., Revotskie, N., and Kagan, A. (1962). *Proc. Roy. Soc. Med.* **55**, 265.
- Day, A. J. (1967). *Advan. Lipid Res.* **5**, 185.
- Day, A. J., and Tume, R. K. (1970). *Atherosclerosis* **11**, 291.
- Day, A. J., and Wahlqvist, M. L. (1968). *Circ. Res.* **23**, 779.
- Day, A. J., and Wahlqvist, M. L. (1970). *Exp. Mol. Pathol.* **13**, 199.
- Day, A. J., and Wilkinson, G. K. (1967). *Circ. Res.* **21**, 593.
- Day, A. J., Wahlqvist, M. L., and Campbell, D. J. (1970). *Atherosclerosis* **11**, 301.
- Dayton, S., and Hashimoto, S. (1966). *Circ. Res.* **19**, 1041.
- Dayton, S., and Hashimoto, S. (1970). *Exp. Mol. Pathol.* **13**, 253.

- Eckles, N. E., Taylor, C. B., Campbell, D. J., and Gould, R. G. (1955). *J. Lab. Clin. Med.* **46**, 359.
- Eisenberg, S., Stein, Y., and Stein, O. (1969). *J. Clin. Invest.* **48**, 2320.
- Field, H., Swell, L., Schools, P. E., and Treadwell, C. R. (1960). *Circulation* **22**, 547.
- Fleischer, S., and Rouser, G. (1965). *J. Amer. Oil. Chem. Soc.* **42**, 588.
- Fletcher, R. F. (1973). *Nutr. Metab.* **15**, 97.
- Flore, H., and Sheppard, B. L. (1970). *Proc. Roy. Soc., Ser. B* **174**, 435.
- French, J. E. (1963). In "Biochemical Problems of Lipids" (A. C. Frazer, ed.), B.B.A. Library, Vol. 1, p. 296. Elsevier, Amsterdam.
- French, J. E. (1966). *Int. Rev. Exp. Pathol.* **5**, 253.
- Geer, J. C., and Guidry, M. A. (1964). *Exp. Mol. Pathol.* **3**, 485.
- Geer, J. C., and Haust, M. D. (1972). "Smooth Muscle Cells in Atherosclerosis," Monographs on Atherosclerosis, Vol. 2. Karger, Basel.
- Gerö, S., Gergely, J., Jakab, L., Székely, J., and Virág, S. (1961). *J. Atheroscler. Res.* **1**, 88.
- Glomset, J. A. (1962). *Biochem. Biophys. Acta* **65**, 128.
- Glomset, J. A. (1968). *J. Lipid Res.* **9**, 155.
- Gould, R. G., LeRoy, G. V., Okita, G. T., Kabara, J. J., Keegan, P., and Bergenstal, D. M. (1955). *J. Lab. Clin. Med.* **46**, 372.
- Gould, R. G., Wissler, R. W., and Jones, R. J. (1963). In "The Evolution of the Atherosclerotic Plaque" (R. J. Jones, ed.), p. 302. Univ. of Chicago Press, Chicago, Illinois.
- Grabar, P. (1959). *Methods Biochem. Anal.* **7**, 1.
- Grabar, P., and Williams, C. A. (1953). *Biochim. Biophys. Acta* **10**, 193.
- Graham, J. M., and Green, C. (1967). *Biochem. J.* **103**, 16c.
- Hagerman, J. S., and Gould, R. G. (1951). *Proc. Soc. Exp. Biol. Med.* **78**, 329.
- Hashimoto, S., and Dayton, S. (1966). *J. Atheroscler. Res.* **6**, 580.
- Haust, M. D. (1968). *Progr. Biochem. Pharmacol.* **4**, 429.
- Haust, M. D. (1971). *Hum. Pathol.* **2**, 1.
- Haust, M. D., Wyllie, J. C., and More, R. H. (1964). *Amer. J. Pathol.* **44**, 255.
- Hess, R. (1973). *Atherogenesis: Initiating Factors*, Ciba Found. Symp. N.S. No. 12, p. 59.
- Hess, R., and Staubli, W. (1968). In "Atherosclerosis" (F. G. Schettler and G. S. Boyd, eds.), p. 49. Elsevier, Amsterdam.
- Holman, R. L. (1961). *Amer. J. Clin. Nutr.* **9**, 565.
- Holman, R. L., McGill, H. C., Strong, J. P., and Geer, J. C. (1958). *Amer. J. Pathol.* **34**, 209.
- Howard, C. F., and Portman, O. W. (1966). *Biochim. Biophys. Acta* **125**, 623.
- Insull, W., and Bartch, G. E. (1966). *J. Clin. Invest.* **45**, 513.
- Iverius, P.-H. (1972). *J. Biol. Chem.* **247**, 2607.
- Jepson, E. M., Billimoria, J. D., and MacLagan, N. F. (1965). *Clin. Sci.* **29**, 383.
- Kao, V. C., and Wissler, R. W. (1965). *Exp. Mol. Pathol.* **4**, 465.
- Kayden, H. J., Segal, B. C., and Hus, K. C. (1959). *J. Clin. Invest.* **31**, 1.
- Klimov, A. N., Denisenko, A. D., and Magracheva, E. Ya. (1974). *Atherosclerosis* **19**, 243.
- Knieriem, H. J., Kao, V. C., and Wissler, R. W. (1967). *Arch. Pathol.* **84**, 118.
- Kramsch, D. M., and Hollander, W. (1973). *J. Clin. Invest.* **52**, 236.
- Lang, P. D., and Insull, W. (1970). *J. Clin. Invest.* **49**, 1479.
- Lindner, J. (1972). *Verh. Deut. Ges. Inn. Med.* **78**, 1166.

- Lofland, H. B., and Clarkson, T. B. (1970). *Proc. Soc. Exp. Med. Biol.* 133, 1.
- Lofland, H. B., Moury, D. M., Hoffman, C., and Clarkson, T. B. (1965). *J. Lipid Res.* 6, 112.
- McGill, H. C., ed. (1968). *Lab. Invest.* 18, 463.
- More, R. H., and Haust, M. D. (1957). *Amer. J. Pathol.* 33, 593.
- Newman, H. A. I., and Zilversmit, D. B. (1962). *J. Biol. Chem.* 237, 2078.
- Newman, H. A. I., and Zilversmit, D. B. (1966). *Circ. Res.* 18, 293.
- Ott, H., Lohss, F., and Gergely, J. (1958). *Klin. Wochenschr.* 36, 383.
- Panganamala, R. V., Geer, J. C., Sharma, H. M., and Cornwell, D. G. (1974). *Atherosclerosis* (in press).
- Patelski, J., Waligora, Z., and Szulc, S. (1967). *J. Atheroscler. Res.* 7, 453.
- Paterson, J. C., Cornish, B. R., and Armstrong, E. C. (1956). *Circulation* 13, 224.
- Paterson, J. C., Armstrong, R., and Armstrong, E. C. (1963). *Circulation* 27, 229.
- Peters, T. J., Takano, T., and de Duve, C. (1973). *Atherogenesis: Initiating Factors. Ciba Found. Symp.*, N.S. No. 12, p. 197.
- Portman, O. W. (1970). *Advan. Lipid Res.* 8, 41.
- Portman, O. W., and Alexander, M. (1966). *Arch. Biochem. Biophys.* 117, 357.
- Portman, O. W., and Alexander, M. (1972). *Biochim. Biophys. Acta* 260, 460.
- Rabinowitz, J. L., Myerson, R. M., and Wohl, G. T. (1960). *Proc. Soc. Exp. Biol. Med.* 105, 241.
- Ressler, N. (1960). *Clin. Chim. Acta* 5, 359.
- Robertson, A. L. (1968). *Progr. Biochem. Pharmacol.* 4, 305.
- Roheim, P. S., Haft, D. E., Gidez, L. I., White, A., and Eder, H. A. (1963). *J. Clin. Invest.* 42, 1277.
- St. Clair, R. W., Lofland, H. B., and Clarkson, T. B. (1968). *J. Lipid Res.* 9, 739.
- St. Clair, R. W., Lofland, H. B., and Clarkson, T. B. (1969). *J. Atheroscler. Res.* 10, 193.
- St. Clair, R. W., Lofland, H. B., and Clarkson, T. B. (1970). *Circ. Res.* 27, 213.
- Schoeff, G. I., and French, J. E. (1968). *Proc. Roy. Soc., Ser. B* 169, 153.
- Schwartz, S. M., and Benditt, E. P. (1972). *Amer. J. Pathol.* 66, 241.
- Schwenk, E., and Stevens, D. F. (1960). *Proc. Soc. Exp. Biol. Med.* 103, 614.
- Scott, P. J., and Hurley, P. J. (1970). *Atherosclerosis* 11, 77.
- Small, D. M. (1970). "Surface Chemistry of Biological Systems," p. 55. Plenum, New York.
- Smith, E. B. (1960). *Lancet* i, 799.
- Smith, E. B. (1965a). *J. Atheroscler. Res.* 5, 224.
- Smith, E. B. (1965b). *J. Atheroscler. Res.* 5, 241.
- Smith, E. B. (1971). *Advan. Exp. Med. Biol.* 16A, 81.
- Smith, E. B. (1974). *Advan. Exp. Med. Biol.* 43, 125.
- Smith, E. B., and Slater, R. S. (1970). *Atherosclerosis* 11, 417.
- Smith, E. B., and Slater, R. S. (1971). *Biochem. J.* 123, 39P.
- Smith, E. B., and Slater, R. S. (1972a). *Atherosclerosis* 15, 37.
- Smith, E. B., and Slater, R. S. (1972b). *Lancet* i, 463.
- Smith, E. B., and Slater, R. S. (1973a). *Atherogenesis: Initiating Factors, Ciba Found. Symp.* N.S. No. 12, p. 39.
- Smith, E. B., and Slater, R. S. (1973b). *Nutr. Metab.* 15, 17.
- Smith, E. B., Evans, P. H., and Downham, M. (1967). *J. Atheroscler. Res.* 7, 171.
- Smith, E. B., Slater, R. S., and Chu, P. K. (1968). *J. Atheroscler. Res.* 8, 399.
- Smith, E. B., Slater, R. S., and Hunter, J. N. (1973). *Atherosclerosis* 18, 479.

- Srinivasan, S. R., Lopez-S, A., Radhakrishnamurthy, B., and Berenson, G. S. (1970). *Atherosclerosis* 12, 321.
- Srinivasan, S. R., Dolan, P., Radhakrishnamurthy, B., and Berenson, G. S. (1972). *Atherosclerosis* 16, 95.
- Steele, J. M., and Kayden, H. J. (1955). *Trans. Ass. Amer. Physicians* 66, 249.
- Stein, Y., and Stein, O. (1973). *Atherogenesis: Initiating Factors, Ciba Found. Symp. N.S. No. 12*, p. 165.
- Taylor, H. E. (1953). *Amer. J. Pathol.* 29, 871.
- Tracy, R. E., Merchant, E. B., and Kao, V. C. (1961). *Circ. Res.* 9, 472.
- Walton, K. W., and Williamson, N. (1968). *J. Atheroscler. Res.* 8, 599.
- Watts, H. F. (1963). In "The Evolution of the Atherosclerotic Plaque" (R. J. Jones, ed.), p. 117. Univ. of Chicago Press, Chicago, Illinois.
- Weinhouse, S., and Hirsch, S. (1940). *Arch. Pathol.* 29, 31.
- Whereat, A. F. (1971). *Advan. Lipid. Res.* 9, 119.
- Wilson, J. D. (1970). *J. Clin. Invest.* 49, 655.
- Wissler, R. W., and Vesselinovitch, D. (1968). *Advan. Lipid Res.* 6, 181.
- Woolf, N., and Pilkington, T. R. E. (1965). *J. Pathol. Bacteriol.* 90, 459.
- Zee, P. (1968). *Pediatrics* 41, 640.
- Zemplényi, T. (1964). *Advan. Lipid Res.* 2, 235.
- Zöllner, N., Wolfram, G., Londong, W., and Kirsch, K. (1966). *Klin. Wochenschr.* 44, 380.